



Module Manual

Master of Science (M.Sc.)

Information and Communication Systems

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Program description

Content

The International Master Program Information and Communication Systems is positioned at the intersection between the classical disciplines of computer science, telecommunication, and networking. In this respect, the program addresses this rapidly evolving area by laying in-depth foundations for the design, analysis and implementation of networked Cyber Physical Systems, as well as the software applications and services running on top of them.

The program is organized over two-years (four semesters), starting on 1st of October each year. It includes around two semesters of lectures and practical courses and almost two semesters devoted to work in a research team (project work) and to the preparation of a master's thesis. The "Master of Science" degree will be awarded. The language of the program is English.

Graduates of the program are provided with the basics and knowledge that are required for a successful engineering activity in the field of information and communication technology, and are prepared to operate in an international environment. They acquire extensive knowledge in the mathematical, engineering and scientific principles of this discipline based on a solid theoretical foundation including all the essential application-oriented aspects. Graduates are qualified to independently resolve problems in the field of information and communications technology and related disciplines.

The graduates are able to apply methods and procedures required to work on technical issues, as well as critically examine new insights to further develop and incorporate them in their work. In this way, they are qualified to carry out their duties in the context of societal responsibly.

Career prospects

The study of Information and Communication Systems provides in-depth training in the areas of Information and Communication Technology, Software Systems, IT Security and Signal Processing. Our graduates get the opportunity to find excellent employment opportunities in the German and international job market, as system developers in the areas such as IT sector, automotive industry, computer gaming, processing or mechanical engineering. This program enables excellent career prospects both in the industrial as well as on the academic job market where a Master's degree qualifies graduates for doctoral studies.

Learning target

Knowledge

The graduates have gained common knowledge from the core qualification and more specific knowledge depending on the selected specialization.

W1) All graduates are able to describe information theory and coding basics.

Specialization Communication Systems:

Graduates can

- **W2C)** show their profound knowledge in digital communications,
- **W3C)** describe their specialized knowledge in communication networks,
- **W4C)** explain software development principles,
- **W5C)** explain signal processing fundamentals.

Specialization Secure and Dependable IT Systems:

Graduates can

- **W2S)** give an overview of software verification,
- **W3S)** describe security principles for information and communication systems,
- **W4S)** explain their specialized knowledge in communication networks,
- **W5S)** describe software development and signal processing principles.

Skills

The ability to apply knowledge in order to perform tasks and solve problems will be supported in the MPICS program. Information and Communication Systems graduates are capable to

- **F1)** solve problems in information and communication systems by applying and adapting techniques, procedures and methods that are required for a successful professional activity and by using engineering systematics,
- **F2)** organize the planning of theoretical and experimental studies in order to develop optimal solutions for complex applications in information and communication technology and evaluate the solutions,
- **F3)** evaluate reliability of developed systems, prepare and review results of practical applications so that they can be used for systems optimization
- **F4)** Investigate, evaluate and integrate new technologies, systems, architecture, services and applications for information and communication systems,
- **F5)** analyse problems using scientific systematics and solve them most effectively to develop economically viable approaches for products and systematically reflect non-technical implications of engineering activity to responsibly involve them in their actions.

Social skills

- **S1)** present and argue the results of their work in written and oral form in an comprehensible way,
- **S2)** communicate and collaborate with international professionals, also of other disciplines,
- **S3)** collaborate in challenging projects of information and communications technology in a responsible position,
- **S4)** develop ideas and solutions in team work.

Autonomy

The MPICS program helps to improve ability and readiness to act independently and responsibly, reflect on own actions and the actions of others, and to develop the own functioning. Information and Communication Systems graduates are capable to

- **A1)** identify knowledge gaps and propose solutions to overcome these gaps,
- **A2)** expand and deepen their knowledge and skills independently, taking into account ecological and economic demands responsibly,
- **A3)** familiarize themselves with complex tasks, define new tasks and develop the necessary knowledge for solving it and to systematically apply appropriate means.

Program structure

The four-semester program is designed modularly and is based on the university-wide standardized program structure with uniform module sizes, i.e., multiples of 6 credit points (CP). The program is organized as follows:

1) Core qualification (42 CP), which consists of:

- the course Information Theory and Coding (6 CP),
- Technical Complementary courses (12 CP),
- Business & Management courses from a catalog (6 CP),
- Non-technical Complementary courses from a catalog (6 CP),
- a Research Project (12 CP). The research project consists of a scientific thesis with documentation.

2) Specialization. The students choose between two specializations (48 CP each):

- Communication Systems, which contains communications, software, and signal processing
- Secure and Dependable IT Systems, which contains cyber-security, networks, software and signal processing

3) Thesis. Students write a master thesis in the 4th semester (30 CP).

The sum of required credit points of this master program is 120 CP.

Core Qualification

Module M0523: Business & Management

Module Responsible	Prof. Matthias Meyer
Admission Requirements	Successful completion of the modul "Foundations of Management"
Recommended Previous Knowledge	None
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence <i>Knowledge</i> - Students are able to find their way around selected special areas of management within the scope of business management. - Students are able to explain basic theories, categories, and models in selected special areas of business management. - Students are able to interrelate technical and management knowledge. <i>Skills</i> - Students are able to apply basic methods in selected areas of business management. - Students are able to explain and give reasons for decision proposals on practical issues in areas of business management. Personal Competence <i>Social Competence</i> - Students are able to communicate in small interdisciplinary groups and to jointly develop solutions for complex problems <i>Autonomy</i> - Students are capable of acquiring necessary knowledge independently by means of research and preparation of material.	
Workload in Hours	Depends on choice of courses
Credit points	6

Courses

Information regarding lectures and courses can be found in the corresponding module handbook published separately.

Module M0524: Non-technical Courses for Master	
Module Responsible	Dagmar Richter
Admission Requirements	None
Recommended Previous Knowledge	None
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence <i>Knowledge</i>	The Nontechnical Academic Programms (NTA) imparts skills that, in view of the TUHH's training profile, professional engineering studies require but are not able to cover fully. Self-reliance, self-management, collaboration and professional and personnel management competences. The department implements these training objectives in its teaching architecture, in its teaching and learning arrangements, in teaching areas and by means of teaching offerings in which students can qualify by opting for specific competences and a competence level at the Bachelor's or Master's level. The teaching offerings are pooled in two different catalogues for nontechnical complementary courses. The Learning Architecture consists of a cross-disciplinarily study offering. The centrally designed teaching offering ensures that courses in the nontechnical academic programms follow the specific profiling of TUHH degree courses. The learning architecture demands and trains independent educational planning as regards the individual development of competences. It also provides orientation knowledge in the form of "profiles". The subjects that can be studied in parallel throughout the student's entire study program - if need be, it can be studied in one to two semesters. In view of the adaptation problems that individuals commonly face in their first semesters after making the transition from school to university and in order to encourage individually planned semesters abroad, there is no obligation to study these subjects in one or two specific semesters during the course of studies. Teaching and Learning Arrangements provide for students, separated into B.Sc. and M.Sc., to learn with and from each other across semesters. The challenge of dealing with interdisciplinarity and a variety of stages of learning in courses are part of the learning architecture and are deliberately encouraged in specific courses. Fields of Teaching are based on research findings from the academic disciplines cultural studies, social studies, arts, historical studies, communication studies, migration studies and sustainability research, and from engineering didactics. In addition, from the winter semester 2014/15 students on all Bachelor's courses will have the opportunity to learn about business management and start-ups in a goal-oriented way. The fields of teaching are augmented by soft skills offers and a foreign language offer. Here, the focus is on encouraging goal-oriented communication skills, e.g. the skills required by outgoing engineers in international and intercultural situations. The Competence Level of the courses offered in this area is different as regards the basic training objective in the Bachelor's and Master's fields. These differences are reflected in the practical examples used, in content topics that refer to different professional application contexts, and in the higher scientific and theoretical level of abstraction in the B.Sc. This is also reflected in the different quality of soft skills, which relate to the different team positions and different group leadership functions of Bachelor's and Master's graduates in their future working life. Specialized Competence (Knowledge) Students can • explain specialized areas in context of the relevant non-technical disciplines, • outline basic theories, categories, terminology, models, concepts or artistic techniques in the disciplines represented in the learning area, • different specialist disciplines relate to their own discipline and differentiate it as well as make connections, • sketch the basic outlines of how scientific disciplines, paradigms, models, instruments, methods and forms of representation in the specialized sciences are subject to individual and socio-cultural interpretation and historicity, • Can communicate in a foreign language in a manner appropriate to the subject.
<i>Skills</i>	Professional Competence (Skills) In selected sub-areas students can • apply basic and specific methods of the said scientific disciplines, • question a specific technical phenomena, models, theories from the viewpoint of another, aforementioned specialist discipline, • to handle simple and advanced questions in aforementioned scientific disciplines in a successful manner, • justify their decisions on forms of organization and application in practical questions in contexts that go beyond the technical relationship to the subject.
Personal Competence <i>Social Competence</i>	Personal Competences (Social Skills)

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<i>Autonomy</i>	Students will be able • to learn to collaborate in different manner, • to present and analyze problems in the abovementioned fields in a partner or group situation in a manner appropriate to the addressees, • to express themselves competently, in a culturally appropriate and gender-sensitive manner in the language of the country (as far as this study-focus would be chosen), • to explain nontechnical items to auditorium with technical background knowledge. Personal Competences (Self-reliance) Students are able in selected areas • to reflect on their own profession and professionalism in the context of real-life fields of application • to organize themselves and their own learning processes • to reflect and decide questions in front of a broad education background • to communicate a nontechnical item in a competent way in written form or verbally • to organize themselves as an entrepreneurial subject country (as far as this study-focus would be chosen)
Workload in Hours	Depends on choice of courses
Credit points	6

Courses

Information regarding lectures and courses can be found in the corresponding module handbook published separately.

Module M0673: Information Theory and Coding			
Courses			
Title	Typ	Hrs/wk	CP
Information Theory and Coding (L0436)	Lecture	3	4
Information Theory and Coding (L0438)	Recitation Section (large)	2	2
Module Responsible	Prof. Gerhard Bauch		
Admission Requirements	None		
Recommended Previous Knowledge	- Mathematics 1-3 - Probability theory and random processes - Basic knowledge of communications engineering (e.g. from lecture "Fundamentals of Communications and Random Processes")		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i> <i>Skills</i> Personal Competence <i>Social Competence</i> <i>Autonomy</i>	The students know the basic definitions for quantification of information in the sense of information theory. They know Shannon's source coding theorem and channel coding theorem and are able to determine theoretical limits of data compression and error-free data transmission over noisy channels. They understand the principles of source coding as well as error-detecting and error-correcting channel coding. They are familiar with the principles of decoding, in particular with modern methods of iterative decoding. They know fundamental coding schemes, their properties and decoding algorithms. The students are familiar with the contents of lecture and tutorials. They can explain and apply them to new problems. The students are able to determine the limits of data compression as well as of data transmission through noisy channels and based on those limits to design basic parameters of a transmission scheme. They can estimate the parameters of an error-detecting or error-correcting channel coding scheme for achieving certain performance targets. They are able to compare the properties of basic channel coding and decoding schemes regarding error correction capabilities, decoding delay, decoding complexity and to decide for a suitable method. They are capable of implementing basic coding and decoding schemes in software. The students can jointly solve specific problems. The students are able to acquire relevant information from appropriate literature sources. They can control their level of knowledge during the lecture period by solving tutorial problems, software tools, clicker system.		
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	90 min		
Assignment for the Following Curricula	Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Information and Communication Systems: Core Qualification: Compulsory International Management and Engineering: Specialisation II. Electrical Engineering: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory		

Course L0436: Information Theory and Coding	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	SoSe
Content	- Introduction to information theory and coding - Definitions of information: Self information, entropy - Binary entropy function - Source coding theorem - Entropy of continuous random variables: Differential entropy, differential entropy of uniformly and Gaussian distributed random variables - Source coding - Principles of lossless source coding - Optimal source codes - Prefix codes, prefix-free codes, instantaneous codes - Morse code - Huffman code - Shannon code

- Bounds on the average codeword length
- Relative entropy, Kullback-Leibler distance, Kullback-Leibler divergence
- Cross entropy
- Lempel-Ziv algorithm
- Lempel-Ziv-Welch (LZW) algorithm
- Text compression and image compression using variants of the Lempel-Ziv algorithm
- Channel models
 - AWGN channel
 - Binary-input AWGN channel
 - Binary symmetric channel (BSC)
 - Relationship between AWGN channel and BSC
 - Binary error and erasure channel (BEEC)
 - Binary erasure channel (BEC)
 - Discrete memoryless channels (DMC)
- Definitions of information for multiple random variables
 - Mutual information and channel capacity
 - Entropy, conditional entropy
 - Chain rules for entropy and mutual information
- Channel coding theorem
- Channel capacity of fundamental channels: BSC, BEC, AWGN channel, binary-input AWGN channel etc.
- Power-limited vs. bandlimited transmission
- Capacity of parallel AWGN channels
 - Waterfilling
 - Examples: Multiple input multiple output (MIMO) channels, complex equivalent baseband channels, orthogonal frequency division multiplex (OFDM)
- Source-channel coding theorem, separation theorem
- Multiuser information theory
 - Multiple access channel (MAC)
 - Broadcast channel
 - Principles of multiple access, time division multiple access (TDMA), frequency division multiple access (FDMA), non-orthogonal multiple access (NOMA), hybrid multiple access
 - Achievable rate regions of TDMA and FDMA with power constraint, energy constraint, power spectral density constraint, respectively
 - Achievable rate region of the two-user and K-user multiple access channels
 - Achievable rate region of the two-user and K user broadcast channels
 - Multiuser diversity
- Channel coding
 - Principles and types of channel coding
 - Code rate, data rate, Hamming distance, minimum Hamming distance, Hamming weight, minimum Hamming weight
 - Error detecting and error correcting codes
 - Simple block codes: Repetition codes, single parity check codes, Hamming code, etc.
 - Syndrome decoding
 - Representations of binary data
 - Non-binary symbol alphabets and non-binary codes
 - Code and encoder, systematic and non-systematic encoders
 - Properties of Hamming distance and Hamming weight
 - Decoding spheres
 - Perfect codes
 - Linear codes
 - Decoding principles
 - Syndrome decoding
 - Maximum a posteriori probability (MAP) decoding and maximum likelihood (ML) decoding
 - Hard decision and soft decision decoding
 - Log-likelihood ratios (LLRs), boxplus operation
 - MAP and ML decoding using log-likelihood ratios
 - Soft-in soft-out decoders
 - Error rate performance comparison of codes in terms of SNR per info bit vs. SNR per code bit
 - Linear block codes
 - Generator matrix and parity check matrix, properties of generator matrix and parity check matrix
 - Dual codes
 - Low density parity check (LDPC) codes
 - Sparse parity check matrix
 - Tanner graphs, cycles and girth
 - Degree distributions
 - Code rate and degree distribution
 - Regular and irregular LDPC codes
 - Message passing decoding
 - Message passing decoding in binary erasure channels (BEC)
 - Systematic encoding using erasure message passing decoding
 - Message passing decoding in binary symmetric channels (BSC)
 - Extrinsic information
 - Bit-flipping decoding
 - Effects of short cycles in the Tanner graph
 - Alternative bit-flipping decoding
 - Soft decision message passing decoding: Sum product decoding

	■ Bit error rate performance of LDPC codes ■ Repeat accumulate codes and variants of repeat accumulate codes ■ Message passing decoding and turbo decoding of repeat accumulate codes ○ Convolutional codes ■ Encoding using shift registers ■ Trellis representation ■ Hard decision and soft decision Viterbi decoding ■ Bit error rate performance of convolutional codes ■ Asymptotic coding gain ■ Viterbi decoding complexity ■ Free distance and optimum convolutional codes ■ Generator polynomial description and octal description ■ Catastrophic convolutional codes ■ Non-systematic and recursive systematic convolutional (RSC) encoders ■ Rate compatible punctured convolutional (RCPC) codes ■ Hybrid automatic repeat request (HARQ) with incremental redundancy ■ Unequal error protection with punctured convolutional codes ■ Error patterns of convolutional codes ○ Concatenated codes ■ Serial concatenated codes ■ Parallel concatenated codes, Turbo codes ■ Iterative decoding, turbo decoding ■ Bit error rate performance of turbo codes ■ Interleaver design for turbo codes ○ Coded modulation ■ Principle of coded modulation ■ Achievable rates with PSK/QAM modulation ■ Trellis coded modulation (TCM) ■ Set partitioning ■ Ungerböck codes ■ Multilevel coding ■ Bit-interleaved coded modulation
Literature	Bossert, M.: Kanalcodierung. Oldenbourg. Friedrichs, B.: Kanalcodierung. Springer. Lin, S., Costello, D.: Error Control Coding. Prentice Hall. Roth, R.: Introduction to Coding Theory. Johnson, S.: Iterative Error Correction. Cambridge. Richardson, T., Urbanke, R.: Modern Coding Theory. Cambridge University Press. Gallager, R. G.: Information theory and reliable communication. Wiley-VCH Cover, T., Thomas, J.: Elements of information theory. Wiley.

Course L0438: Information Theory and Coding	
Typ	Recitation Section (large)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1246: Technical Complementary Course for IMPICS (according to Subject Specific Regulations)			
Courses			
Title	Typ	Hrs/wk	CP
Module Responsible	Prof. Riccardo Scandariato		
Admission Requirements	None		
Recommended Previous Knowledge			
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i> <i>Skills</i> Personal Competence <i>Social Competence</i> <i>Autonomy</i>			
Workload in Hours	Depends on choice of courses		
Credit points	12		
Assignment for the Following Curricula	Information and Communication Systems: Core Qualification: Compulsory		

Module M1776: Research Project ICS			
Courses			
Title Research Project ICS (L2919)	Typ Projection Course	Hrs/wk 8	CP 12
Module Responsible	Prof. Riccardo Scandariato		
Admission Requirements	None		
Recommended Previous Knowledge	Basic knowledge and techniques in the chosen field of specialization.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	Students are able to acquire advanced knowledge in a specific field of Computer Science or a closely related subject. Students are able to work self-dependent in a field of Computer Science or a closely related field.		
<i>Skills</i>			
Personal Competence <i>Social Competence</i>			
<i>Autonomy</i>			
Workload in Hours	Independent Study Time 248, Study Time in Lecture 112		
Credit points	12		
Course achievement	None		
Examination	Study work		
Examination duration and scale	Presentation of a current research topic (25-30 min and 5 min discussion)		
Assignment for the Following Curricula	Information and Communication Systems: Core Qualification: Compulsory		

Course L2919: Research Project ICS	
Typ	Projection Course
Hrs/wk	8
CP	12
Workload in Hours	Independent Study Time 248, Study Time in Lecture 112
Lecturer	Dozenten des SD E
Language	EN
Cycle	WiSe
Content	Current research topics of the chosen specialization.
Literature	Aktuelle Literatur zu Forschungsthemen aus der gewählten Vertiefungsrichtung. / Current literature on research topics of the chosen specialization.

Specialization Communication Systems

Graduates of the Communication Systems specialisation are qualified to independently resolve problems in communication networks and digital communications. They also have profound knowledge in software development principles and signal processing. Graduates are qualified to independently resolve problems in communication systems technology and related disciplines.

The Communication Systems specialisation is recommended for students who already bring along a good mathematical foundation, basic knowledge in computer science and/or electrical engineering with focus on information and communication technology.

Module M0676: Digital Communications				
Courses				
Title		Typ	Hrs/wk	CP
Digital Communications (L0444)		Lecture	2	3
Digital Communications (L0445)		Recitation Section (large)	2	2
Laboratory Digital Communications (L0646)		Practical Course	1	1
Module Responsible	Prof. Gerhard Bauch			
Admission Requirements	None			
Recommended Previous Knowledge	- Mathematics 1-3 - Signals and Systems - Fundamentals of Communications and Random Processes			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students are able to understand, compare and design modern digital information transmission schemes. They are familiar with the properties of linear and non-linear digital modulation methods. They can describe distortions caused by transmission channels and design and evaluate detectors including channel estimation and equalization. They know the principles of single carrier transmission and multi-carrier transmission as well as the fundamentals of basic multiple access schemes. The students are familiar with the contents of lecture and tutorials. They can explain and apply them to new problems. <i>Skills</i> The students are able to design and analyse a digital information transmission scheme including multiple access. They are able to choose a digital modulation scheme taking into account transmission rate, required bandwidth, error probability, and further signal properties. They can design an appropriate detector including channel estimation and equalization taking into account performance and complexity properties of suboptimum solutions. They are able to set parameters of a single carrier or multi carrier transmission scheme and trade the properties of both approaches against each other. <i>Personal Competence</i> <i>Social Competence</i> The students can jointly solve specific problems. <i>Autonomy</i> The students are able to acquire relevant information from appropriate literature sources. They can control their level of knowledge during the lecture period by solving tutorial problems, software tools, clicker system.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Written elaboration	
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Data Science: Specialisation II. Computer Science: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Electrical Engineering and Information Technology: Core Qualification: Compulsory Electrical Engineering: Core Qualification: Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory International Management and Engineering: Specialisation II. Electrical Engineering: Elective Compulsory Microelectronics and Microsystems: Core Qualification: Elective Compulsory			

Course L0444: Digital Communications	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	WiSe
Content	- Repetition: Baseband Transmission - Pulse shaping: Non-return to zero (NRZ) rectangular pulses, raised-cosine pulses, square-root raised-cosine pulses - Power spectral density (psd) of baseband signals - Intersymbol interference (ISI) - First and second Nyquist criterion - AWGN channel

- Matched filter
 - Matched-filter receiver and correlation receiver
 - Noise whitening matched filter
 - Discrete-time AWGN channel model
- Representation of bandpass signals and systems in the equivalent baseband
 - Quadrature amplitude modulation (QAM)
 - Equivalent baseband signal and system
 - Analytical signal
 - Equivalent baseband random process, equivalent baseband white Gaussian noise process
 - Equivalent baseband AWGN channel
 - Equivalent baseband channel model with frequency-offset and phase noise
 - Equivalent baseband Rayleigh fading and Rice fading channel models
 - Equivalent baseband frequency-selective channel model
 - Discrete memoryless channels (DMC)
- Bandpass transmission via carrier modulation
 - Amplitude modulation, frequency modulation, phase modulation
 - Linear digital modulation methods
 - On-off keying, M-ary amplitude shift keying (M-ASK), M-ary phase shift keying (M-PSK), M-ary quadrature amplitude modulation (M-QAM), offset-QPSK
 - Signal space representation of transmit signal constellations and signals
 - Energy of linear digital modulated signals, average energy per symbol
 - Power spectral density of linear digital modulated signals
 - Bandwidth efficiency
 - Correlation coefficient of elementary signals
 - Error probabilities of linear digital modulation methods
 - Error functions
 - Gray mapping and natural mapping
 - Bit error probabilities, symbol error probabilities, pairwise symbol error probabilities
 - Euclidean distance and Hamming distance
 - Exact and approximate computation of error probabilities
 - Performance comparison of modulation schemes in terms of per bit SNR vs. per symbol SNR
 - Hierarchical modulation, multilevel modulation
 - Effects of carrier phase offset and carrier frequency offset
 - Differential modulation
 - M-ary differential phase shift keying (M-PSK)
 - Coherent and non-coherent detection of DPSK
 - p/M-differential phase shift keying (p/M-DPSK)
 - Differential amplitude and phase shift keying (DAPSK)
 - Non-linear digital modulation methods
 - Frequency shift keying (FSK)
 - Modulation index
 - Minimum shift keying (MSK)
 - Offset-QPSK representation of MSK
 - MSK with differential precoding and rotation
 - Bit error probabilities of MSK
 - Gaussian minimum shift keying (GMSK)
 - Power spectral density of MSK and GMSK
 - Continuous phase modulation (CPM)
 - General description of CPM signals
 - Frequency pulses and phase pulses
 - Coherent and non-coherent detection of FSK
 - Performance comparison of linear and non-linear digital modulation methods
- Frequency-selective channels, ISI channels
 - Intersymbol interference and frequency-selectivity
 - RMS delay spread
 - Narrowband and broadband channels
 - Equivalent baseband transmission model for frequency-selective channels
 - Receive filter design
- Equalization
 - Symbol-spaced and fractionally-spaced equalizers
 - Inverse system
 - Non-recursive linear equalizers
 - Linear zero-forcing (ZF) equalizer
 - Linear minimum mean squared error (MMSE) equalizer
 - Non-linear equalization:
 - Decision feedback equalizer (DFE)
 - Tomlinson-Harashima precoding
 - Maximum a posteriori probability (MAP) and maximum likelihood equalizer, Viterbi algorithm
- Single-carrier vs. multi-carrier transmission
- Multi-carrier transmission
 - General multicarrier transmission
 - Orthogonal frequency division multiplex (OFDM)
 - OFDM implementation using the Fast Fourier Transform (FFT)
 - Cyclic guard interval
 - Power spectral density of OFDM

Module Manual M.Sc. "Information and Communication Systems"

	▪ Peak-to-average power ratio (PAPR) • Multiple access ◦ Principles of time division multiple access (TDMA), frequency division multiple access (FDMA), code division multiple access (CDMA), non-orthogonal multiple access (NOMA), hybrid multiple access • Spread spectrum communications ◦ Direct sequence spread spectrum communications ◦ Frequency hopping ◦ Protection against eavesdropping ◦ Protection against narrowband jammers ◦ Short vs. long spreading codes ◦ Direct sequence spread spectrum communications in frequency-selective channels ▪ Rake receiver ◦ Code division multiple access (CDMA) ▪ Design criteria of spreading sequences, autocorrelation function and crosscorrelation function of spreading sequences ▪ Intersymbol interference (ISI) and multiple access interference (MAI) ▪ Pseudo noise (PN) sequences, maximum length sequences (m-sequences), Gold codes, Walsh-Hadamard codes, orthogonal variable spreading factor (OVSF) codes ▪ Multicode transmission ▪ CDMA in uplink and downlink of a wireless communications system ▪ Single-user detection vs. multi-user detection
Literature	K. Kammeyer: Nachrichtenübertragung, Teubner P.A. Höher: Grundlagen der digitalen Informationsübertragung, Teubner. J.G. Proakis, M. Salehi: Digital Communications. McGraw-Hill. S. Haykin: Communication Systems. Wiley R.G. Gallager: Principles of Digital Communication. Cambridge A. Goldsmith: Wireless Communication. Cambridge. D. Tse, P. Viswanath: Fundamentals of Wireless Communication. Cambridge.

Course L0445: Digital Communications	
Typ	Recitation Section (large)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L0646: Laboratory Digital Communications	
Typ	Practical Course
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	WiSe
Content	- DSL transmission - Random processes - Digital data transmission
Literature	K. Kammeyer: Nachrichtenübertragung, Teubner P.A. Höher: Grundlagen der digitalen Informationsübertragung, Teubner. J.G. Proakis, M. Salehi: Digital Communications. McGraw-Hill. S. Haykin: Communication Systems. Wiley R.G. Gallager: Principles of Digital Communication. Cambridge A. Goldsmith: Wireless Communication. Cambridge. D. Tse, P. Viswanath: Fundamentals of Wireless Communication. Cambridge.

Module M0836: Communication Networks								
Courses								
Title		Typ	Hrs/wk	CP				
Selected Topics of Communication Networks (L0899)		Project-/problem-based Learning	2	2				
Communication Networks (L0897)		Lecture	2	2				
Communication Networks Exercise (L0898)		Project-/problem-based Learning	1	2				
Module Responsible	Dr. Koojana Kuladinithi							
Admission Requirements	None							
Recommended Previous Knowledge	- Fundamental stochastics - Basic understanding of computer networks and/or communication technologies is beneficial							
Educational Objectives	After taking part successfully, students have reached the following learning results							
Professional Competence	Knowledge Students are able to describe the principles and structures of communication networks in detail. They can explain the formal description methods of communication networks and their protocols. They are able to explain how current and complex communication networks work and describe the current research in these examples. Skills Students are able to evaluate the performance of communication networks using the learned methods. They are able to work out problems themselves and apply the learned methods. They can apply what they have learned autonomously on further and new communication networks. Personal Competence Social Competence Students are able to define tasks themselves in small teams and solve these problems together using the learned methods. They can present the obtained results. They are able to discuss and critically analyse the solutions. Autonomy Students are able to obtain the necessary expert knowledge for understanding the functionality and performance capabilities of new communication networks independently.							
Workload in Hours					Independent Study Time 110, Study Time in Lecture 70			
Credit points					6			
Course achievement					None			
Examination					Presentation			
Examination duration and scale	1.5 hours colloquium with three students, therefore about 30 min per student. Topics of the colloquium are the posters from the previous poster session and the topics of the module.							
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory							

Course L0899: Selected Topics of Communication Networks	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Koojana Kuladinithi
Language	EN
Cycle	WiSe
Content	Example networks selected by the students will be researched on in a PBL course by the students in groups and will be presented in a poster session at the end of the term.
Literature	see lecture

Course L0897: Communication Networks	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Koojana Kuladinithi
Language	EN
Cycle	WiSe
Content	consisting of 3 modules. - Lecture on the analysis of communication networks - Introduction to Communication Networks - Evolution of PSTN and Blocking Probability - TCP/IP Based Networks and Mobile Cellular Networks - Point-to-Point Communication - Internet - Part 1: Network Layer & Part 2: Transport Layer - Local Area Networks - Ethernet - Wireless Local Area Networks - PBL Exercises of communication networks - Performance Metrics - Trial & Error vs. Engineering Approach - Reliable Medium Access Control - Mini poster: Fast and Reliable End-to-End Data Transmission - Selected topics in communication networks This is organised as a Poster presentation of the PBL5 done for the examination. The students are asked to work on a selected topic in communication networks. This topic is selected based on the current related topic.
Literature	The students should have the basic knowledge of communication networks. Lecture 1 explains some terminologies in brief. The following book references are recommended. - Keith W. Ross James F. Kurose. Computer Networking - A Top-Down Approach. 7th ed. Pearson, 2017. A. S. Tanenbaum. Computer Networks. 5th ed. Prentice Hall, 2011. - Schiller: Mobile Communications, 8th edition. - Internet, Wikipedia

Course L0898: Communication Networks Exercise	
Typ	Project-/problem-based Learning
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Dr. Koojana Kuladinithi
Language	EN
Cycle	WiSe
Content	Part of the content of the lecture Communication Networks are reflected in computing tasks in groups, others are motivated and addressed in the form of a PBL exercise.
Literature	announced during lecture

Module M0710: Microwave Engineering				
Courses				
Title	Typ		Hrs/wk	CP
Microwave Engineering (L0573)	Lecture		2	3
Microwave Engineering (L0574)	Recitation Section (large)		2	2
Microwave Engineering (L0575)	Practical Course		1	1
Module Responsible	Prof. Alexander Kölpin			
Admission Requirements	None			
Recommended Previous Knowledge	Fundamentals of communication engineering, semiconductor devices and circuits. Basics of Wave propagation from transmission line theory and theoretical electrical engineering.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	Students can explain the propagation of electromagnetic waves and related phenomena. They can describe transmission systems and components. They can name different types of antennas and describe the main characteristics of antennas. They can explain noise in linear circuits, compare different circuits using characteristic numbers and select the best one for specific scenarios.			
<i>Skills</i>	Students are able to calculate the propagation of electromagnetic waves. They can analyze complete transmission systems und configure simple receiver circuits. They can calculate the characteristic of simple antennas and arrays based on the geometry. They can calculate the noise of receivers and the signal-to-noise-ratio of transmission systems. They can apply their theoretical knowledge to the practical courses.			
Personal Competence				
<i>Social Competence</i>	Students work together in small groups during the practical courses. Together they document, evaluate and discuss their results.			
<i>Autonomy</i>	Students are able to relate the knowledge gained in the course to contents of previous lectures. With given instructions they can extract data needed to solve specific problems from external sources. They are able to apply their knowledge to the laboratory courses using the given instructions.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Subject	theoretical and practical work
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Core Qualification: Compulsory Electrical Engineering: Core Qualification: Compulsory Information and Communication Systems: Specialisation Communication Systems: Elective Compulsory International Management and Engineering: Specialisation II. Electrical Engineering: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory Microelectronics and Microsystems: Specialisation Engineering for Quantum Technologies: Elective Compulsory			

Course L0573: Microwave Engineering	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Alexander Kölpin
Language	EN
Cycle	WiSe
Content	- Antennas: Analysis - Characteristics - Realizations - Radio Wave Propagation - Transmitter: Power Generation with Vacuum Tubes and Transistors - Receiver: Preamplifier - Heterodyning - Noise - Selected System Applications
Literature	H.-G. Unger, „Elektromagnetische Theorie für die Hochfrequenztechnik, Teil I“, Hüthig, Heidelberg, 1988 H.-G. Unger, „Hochfrequenztechnik in Funk und Radar“, Teubner, Stuttgart, 1994 E. Voges, „Hochfrequenztechnik - Teil II: Leistungsrohren, Antennen und Funkübertragung, Funk- und Radartechnik“, Hüthig, Heidelberg, 1991 E. Voges, „Hochfrequenztechnik“, Hüthig, Bonn, 2004 C.A. Balanis, „Antenna Theory“, John Wiley and Sons, 1982 R. E. Collin, „Foundations for Microwave Engineering“, McGraw-Hill, 1992 D. M. Pozar, „Microwave and RF Design of Wireless Systems“, John Wiley and Sons, 2001 D. M. Pozar, „Microwave Engineerin“, John Wiley and Sons, 2005

Course L0574: Microwave Engineering	
Typ	Recitation Section (large)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Alexander Kölpin
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L0575: Microwave Engineering	
Typ	Practical Course
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Alexander Kölpin
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0637: Advanced Concepts of Wireless Communications			
Courses			
Title	Typ	Hrs/wk	CP
Advanced Concepts of Wireless Communications (L0297)	Lecture	3	4
Advanced Concepts of Wireless Communications (L0298)	Recitation Section (large)	2	2
Module Responsible	Dr. Rainer Grünheid		
Admission Requirements	None		
Recommended Previous Knowledge	• Lecture "Signals and Systems" • Lecture "Fundamentals of Telecommunications and Stochastic Processes" • Lecture "Digital Communications"		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> Students are able to explain the general as well as advanced principles and techniques that are applied to wireless communications. They understand the properties of wireless channels and the corresponding mathematical description. Furthermore, students are able to explain the physical layer of wireless transmission systems. In this context, they are proficient in the concepts of multicarrier transmission (OFDM), modulation, error control coding, channel estimation and multi-antenna techniques (MIMO). Students can also explain methods of multiple access. On the example of contemporary communication systems (LTE, 5G) they can put the learnt content into a larger context. The students are familiar with the contents of lecture and tutorials. They can explain and apply them to new problems. <i>Skills</i> Using the acquired knowledge, students are able to understand the design of current and future wireless systems. Moreover, given certain constraints, they can choose appropriate parameter settings of communication systems. Students are also able to assess the suitability of technical concepts for a given application. Personal Competence <i>Social Competence</i> Students can jointly elaborate tasks in small groups and present their results in an adequate fashion. <i>Autonomy</i> Students are able to extract necessary information from given literature sources and put it into the perspective of the lecture. They can continuously check their level of expertise with the help of accompanying measures (such as online tests, clicker questions, exercise tasks) and, based on that, to steer their learning process accordingly. They can relate their acquired knowledge to topics of other lectures, e.g., "Fundamentals of Communications and Stochastic Processes" and "Digital Communications".		
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	90 minutes; scope: content of lecture and exercise		
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory		

Course L0297: Advanced Concepts of Wireless Communications	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Dr. Rainer Grünheid
Language	EN
Cycle	SoSe
Content	The lecture deals with technical principles and related concepts of mobile communications. In this context, the main focus is put on the physical and data link layer of the ISO-OSI stack. In the lecture, the transmission medium, i.e., the mobile radio channel, serves as the starting point of all considerations. The characteristics and the mathematical descriptions of the radio channel are discussed in detail. Subsequently, various physical layer aspects of wireless transmission are covered, such as channel coding, modulation/demodulation, channel estimation, synchronization, and equalization. Moreover, the different uses of multiple antennas at the transmitter and receiver, known as MIMO techniques, are described. Besides these physical layer topics, concepts of multiple access schemes in a cellular network are outlined. In order to illustrate the above-mentioned technical solutions, the lecture will also provide a system view, highlighting the basics of some contemporary wireless systems, including LTE, LTE Advanced, and 5G New Radio.
Literature	John G. Proakis, Masoud Salehi: Digital Communications. 5th Edition, Irwin/McGraw Hill, 2007 David Tse, Pramod Viswanath: Fundamentals of Wireless Communication. Cambridge, 2005 Bernard Sklar: Digital Communications: Fundamentals and Applications. Second Edition, Pearson, 2013 Stefani Sesia, Issam Toufik, Matthew Baker: LTE - The UMTS Long Term Evolution. Second Edition, Wiley, 2011 Erik Dahlman, Stefan Parkvall, Johan Sköld: 5G NR - The Next Generation Wireless Access Technology. Second Edition, Academic Press, 2021

Course L0298: Advanced Concepts of Wireless Communications	
Typ	Recitation Section (large)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Rainer Grünheid
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0837: Simulation of Communication Networks								
Courses								
Title		Typ	Hrs/wk	CP				
Simulation of Communication Networks (L0887)		Project-/problem-based Learning	3	4				
Simulation of Communication Networks (L3476)		Lecture	2	2				
Module Responsible	Dr. Koojana Kuladinithi							
Admission Requirements	None							
Recommended Previous Knowledge	• Knowledge of computer and communication networks • Basic programming skills							
Educational Objectives	After taking part successfully, students have reached the following learning results							
Professional Competence	Knowledge Students are able to explain the necessary stochastics, the discrete event simulation technology and modelling of networks for performance evaluation. Skills Students are able to apply the method of simulation for performance evaluation to different, also not practiced, problems of communication networks. The students can analyse the obtained results and explain the effects observed in the network. They are able to question their own results. Personal Competence Social Competence Students are able to acquire expert knowledge in groups, present the results, and discuss solution approaches and results. They are able to work out solutions for new problems in small teams. Autonomy Students are able to transfer independently and in discussion with others the acquired method and expert knowledge to new problems. They can identify missing knowledge and acquire this knowledge independently.							
Workload in Hours					Independent Study Time 110, Study Time in Lecture 70			
Credit points					6			
Course achievement					None			
Examination					Oral exam			
Examination duration and scale	30 min							
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory							

Course L0887: Simulation of Communication Networks	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Dr. Koojana Kuladinithi
Language	EN
Cycle	SoSe
Content	In the course necessary basic stochastics and the discrete event simulation are introduced. Also simulation models for communication networks, for example, traffic models, mobility models and radio channel models are presented in the lecture. Students work with a simulation tool, where they can directly try out the acquired skills, algorithms and models. At the end of the course increasingly complex networks and protocols are considered and their performance is determined by simulation.
Literature	Skript des Instituts für Kommunikationsnetze Further literature is announced at the beginning of the lecture.

Course L3476: Simulation of Communication Networks	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Koojana Kuladinithi
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0638: Modern Wireless Systems				
Courses				
Title	Typ		Hrs/wk	CP
Selected Topics of Modern Wireless Systems (L1982)	Project-/problem-based Learning		2	3
Modern Wireless Systems (L0296)	Lecture		3	3
Module Responsible	Dr. Rainer Grünheid			
Admission Requirements	None			
Recommended Previous Knowledge	- Lecture "Digital Communications" - Lecture "Advanced Concepts of Wireless Communications"			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	Students have an overview of a variety of contemporary wireless systems of different size and complexity. They understand the technical solutions from the perspective of the physical and data link layer. They have developed a system view and are aware of the technical arguments, considering the respective applications and associated constraints. For several examples (e.g., 5G New Radio), students are able to explain different concepts in a very deep technical detail. The students are familiar with the contents of lecture and PBL course. They can explain and apply them to new problems.			
<i>Skills</i>	Students have developed a system view. They can transfer their knowledge to evaluate other systems, not discussed in the lecture, and to understand the respective technical solutions. Given specific constraints and technical requirements, students are in a position to make proposals for certain design aspects by an appropriate assessment and the consideration of alternatives.			
Personal Competence <i>Social Competence</i>	Students can jointly elaborate tasks in small groups and present their results in an adequate fashion.			
<i>Autonomy</i>	Students are able to extract necessary information from given literature sources and put it into the perspective of the lecture. They can continuously check their level of expertise with the help of accompanying measures (such as online tests, clicker questions, exercise tasks) and, based on that, to steer their learning process accordingly. They can relate their acquired knowledge to topics of other lectures, e.g., "Digital Communications" and "Advanced Topics of Wireless Communications".			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Subject	theoretical and PBL-Kurs mit Posterpräsentation
			practical work	
Examination	Oral exam			
Examination duration and scale	40 min			
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Elective Compulsory			

Course L1982: Selected Topics of Modern Wireless Systems	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Rainer Grünheid
Language	EN
Cycle	WiSe
Content	In this course, selected "hot" topics of modern wireless systems will be covered. For that purpose, students work in small groups to elaborate a given subject, including a quantitative analysis with provided simulation tools. The results will be presented in a poster session or a talk towards the end of the semester. Possible topics can include various system concepts and related technical principles, such as: - WLAN systems 5G systems - Millimeter wave communication - Visible light communication - Cooperative Multipoint - Massive MIMO - Massive machine-type communication - Interference cancellation - Non-orthogonal multiple access - Heterogeneous networks ...
Literature	will be provided, depending on the given topics

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Course L0296: Modern Wireless Systems	
Typ	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Dr. Rainer Grünheid
Language	EN
Cycle	WiSe
Content	The lecture gives an overview of contemporary wireless communication concepts and related techniques from a system point of view. For that purpose, different systems, ranging from Wireless Personal to Wide Area Networks, are covered, mainly discussing the physical and data link layer. Systems under consideration include: - Near Field Communication (NFC) - ZigBee / IEEE 802.15.4 - Bluetooth - IEEE 802.11 family - L-band Digital Aeronautical Communication System (LDACS) - Long Term Evolution (LTE) and LTE Advanced - 5G New Radio A special focus is placed on 4th and 5th generation networks; in particular, an in-depth view into the technical principles of the 5G New Radio standard is given.
Literature	John G. Proakis, Masoud Salehi: Digital Communications. 5th Edition, Irwin/McGraw Hill, 2007 Stefani Sesia, Issam Toufik, Matthew Baker: LTE - The UMTS Long Term Evolution. Second Edition, Wiley, 2011 Erik Dahlman, Stefan Parkvall, Johan Sköld: 5G NR - The Next Generation Wireless Access Technology. Second Edition, Academic Press, 2021

Module M2115: Advanced Seminar EIM I					
Courses					
Title		Typ	Hrs/wk	CP	
Advanced Seminar EIM I (L3385)		Seminar	2	3	
Module Responsible	Dozenten des SD E				
Admission Requirements	None				
Recommended Previous Knowledge	Basic knowledge in the area of the seminar within the fields of computer science, communication technology, electrical engineering or mathematics at the Master's level.				
Educational Objectives	After taking part successfully, students have reached the following learning results				
Professional Competence					
Knowledge					The students are able to • explicate a specific topic from the field of the seminar, • describe complex issues, • present different views and evaluate in a critical way.
Skills					The students are able to • familiarize in a specific topic in limited time, • realize a literature survey on the specific topic and cite in a correct way, • elaborate a presentation and give a lecture to a selected audience, • sum up the presentation in 10-15 lines, • answer questions in the final discussion.
Personal Competence					
Social Competence					The students are able to • elaborate and introduce a topic for a certain audience, • discuss the topic, content and structure of the presentation with the instructor, • discuss certain aspects with the audience, and • as the lecturer listen and respond to questions from the audience.
Autonomy	The students are able to • define the task in question in an autonomous way, • develop the necessary knowledge, • use appropriate work equipment, and • guided by an instructor critically check the working status.				
Workload in Hours	Depends on choice of courses				
Credit points	3				
Assignment for the Following Curricula	Computer Science: Specialisation IV. Subject Specific Focus: Elective Compulsory Data Science: Core Qualification: Compulsory Electrical Engineering and Information Technology: Specialisation Microwave Engineering, Optics, and Electromagnetic Compatibility: Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Compulsory Electrical Engineering and Information Technology: Specialisation Nanoelectronics and Microsystems Technology: Compulsory Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Compulsory Computer Science in Engineering: Specialisation IV. Subject Specific Focus: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory				

Course L3385: Advanced Seminar EIM I	
Typ	Seminar
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Fachtheoretisch-fachpraktische Arbeit
Examination duration and scale	x
Lecturer	Dozenten des SD E
Language	EN
Cycle	WiSe/SoSe
Content	- Seminar presentations by enrolled students about selected topics of computer science, communication technology, electrical engineering or mathematics - Active participation in discussions
Literature	Wird vom Veranstalter bekanntgegeben.

Module M2116: Advanced Seminar EIM II				
Courses				
Title		Typ	Hrs/wk	CP
Advanced Seminar EIM II (L3386)		Seminar	2	3
Module Responsible	Dozenten des SD E			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge in the area of the seminar within the fields of computer science, communication technology, electrical engineering or mathematics at the Master's level.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Knowledge The students are able to - explicate a specific topic from the field of the seminar, - describe complex issues, - present different views and evaluate in a critical way. Skills The students are able to - familiarize in a specific topic in limited time, - realize a literature survey on the specific topic and cite in a correct way, - elaborate a presentation and give a lecture to a selected audience, - sum up the presentation in 10-15 lines, - answer questions in the final discussion.			
Personal Competence				
Social Competence				
Autonomy				
Workload in Hours	Depends on choice of courses			
Credit points	3			
Assignment for the Following Curricula	Computer Science: Specialisation IV. Subject Specific Focus: Elective Compulsory Computer Science in Engineering: Specialisation IV. Subject Specific Focus: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory			

Course L3386: Advanced Seminar EIM II	
Typ	Seminar
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Fachtheoretisch-fachpraktische Arbeit
Examination duration and scale	x
Lecturer	Dozenten des SD E
Language	EN
Cycle	WiSe/SoSe
Content	- Seminar presentations by enrolled students about selected topics of computer science, communication technology, electrical engineering or mathematics - Active participation in discussions
Literature	Wird vom Veranstalter bekanntgegeben.

Focus Signal Processing

Module M0677: Digital Signal Processing and Digital Filters

Courses

Title	Typ	Hrs/wk	CP
Digital Signal Processing and Digital Filters (L0446)	Lecture	3	4
Digital Signal Processing and Digital Filters (L0447)	Recitation Section (large)	2	2

Module Responsible	Prof. Gerhard Bauch
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Admission Requirements	None
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Recommended Previous Knowledge	- Mathematics 1-3 - Signals and Systems - Fundamentals of signal and system theory as well as random processes. - Fundamentals of spectral transforms (Fourier series, Fourier transform, Laplace transform)
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Educational Objectives	After taking part successfully, students have reached the following learning results
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Professional Competence	
<i>Knowledge</i>	The students know and understand basic algorithms of digital signal processing. They are familiar with the spectral transforms of discrete-time signals and are able to describe and analyse signals and systems in time and image domain. They know basic structures of digital filters and can identify and assess important properties including stability. They are aware of the effects caused by quantization of filter coefficients and signals. They are familiar with the basics of adaptive filters. They can perform traditional and parametric methods of spectrum estimation, also taking a limited observation window into account. The students are familiar with the contents of lecture and tutorials. They can explain and apply them to new problems.
<i>Skills</i>	The students are able to apply methods of digital signal processing to new problems. They can choose and parameterize suitable filter structures. In particular, they can design adaptive filters according to the minimum mean squared error (MMSE) criterion and develop an efficient implementation, e.g. based on the LMS or RLS algorithm. Furthermore, the students are able to apply methods of spectrum estimation and to take the effects of a limited observation window into account.
Personal Competence	
<i>Social Competence</i>	The students can jointly solve specific problems.
<i>Autonomy</i>	The students are able to acquire relevant information from appropriate literature sources. They can control their level of knowledge during the lecture period by solving tutorial problems, software tools, clicker system.

Workload in Hours	Independent Study Time 110, Study Time in Lecture 70
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Credit points	6
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Course achievement	None
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Examination	Written exam
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Examination duration and scale	90 min
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Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory Mechanical Engineering and Management: Specialisation Mechatronics: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory
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Course L0446: Digital Signal Processing and Digital Filters	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	WiSe
Content	• Transforms of discrete-time signals: ◦ Discrete-time Fourier Transform (DTFT) ◦ Discrete Fourier-Transform (DFT), Fast Fourier Transform (FFT) ◦ Z-Transform • Correspondence of continuous-time and discrete-time signals, sampling, sampling theorem • Fast convolution, Overlap-Add-Method, Overlap-Save-Method • Fundamental structures and basic types of digital filters • Characterization of digital filters using pole-zero plots, important properties of digital filters • Quantization effects • Design of linear-phase filters • Fundamentals of stochastic signal processing and adaptive filters ◦ MMSE criterion ◦ Wiener Filter ◦ LMS- and RLS-algorithm • Traditional and parametric methods of spectrum estimation
Literature	K.-D. Kammeyer, K. Kroschel: Digitale Signalverarbeitung. Vieweg Teubner. V. Oppenheim, R. W. Schaffer, J. R. Buck: Zeitdiskrete Signalverarbeitung. Pearson StudiumA. V. W. Hess: Digitale Filter. Teubner. Oppenheim, R. W. Schaffer: Digital signal processing. Prentice Hall. S. Haykin: Adaptive filter theory. L. B. Jackson: Digital filters and signal processing. Kluwer. T.W. Parks, C.S. Burrus: Digital filter design. Wiley.

Course L0447: Digital Signal Processing and Digital Filters	
Typ	Recitation Section (large)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1749: Energy Efficiency in Embedded Systems			
Courses			
Title	Typ	Hrs/wk	CP
Energy Efficiency in Embedded Systems (L2870)	Lecture	2	3
Energy Efficiency in Embedded Systems (L2872)	Project/problem-based Learning	2	2
Energy Efficiency in Embedded Systems (L2871)	Recitation Section (large)	1	1
Module Responsible	Prof. Ulf Kulau		
Admission Requirements	None		
Recommended Previous Knowledge	• Computer Engineering (mandatory) • Programming Skills in C (mandatory) • Computer Architecture (recommended) ENG: The following books are very good sources for the necessary background knowledge: 1. Harris, David, and N. Weste: CMOS VLSI Design ed., Pearson Education, 2010 2. Rabaey, Jan: Low Power Design Essentials (Integrated Circuits and Systems), Springer, 2009		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	Motivation: In the field of computer science we have only limited possibilities to influence the efficiency of the hardware directly, respectively we are dependent on the manufacturers (e.g. of microcontrollers). However, in order to exploit the full potential of the hardware we are given at the system level, we need a deeper understanding of the background, processes and mechanisms of power dissipation in embedded systems. Where does the power dissipation come from, what happens at the hardware level, what mechanisms can I use directly/indirectly, what is the tradeoff between flexibility and efficiency,..... are only a few questions, which will be elaborated and discussed in this event. Contents of teaching: • Motivation and power dissipation on semiconductor level • Power dissipation of digital circuits, in particular CMOS • Power Management in Hard- and Software (Sleep Modes, DVS, FS, Undervolting) • Energy efficient system design (applications) • Energy Harvesting and Transiently Powered Computing (TPC)		
<i>Skills</i>	Upon completion of this module, students will have a deeper understanding of hardware and software mechanisms for evaluating and developing energy-efficient embedded systems • They have a deeper understanding of the electrotechnical basics of power dissipation in digital systems • They can analyze the power dissipation of systems at any level and apply appropriate methods to increase efficiency • They can use a variety of standard techniques to achieve "Energy Efficiency by Design" • They can model, evaluate as well as implement energy-autonomous systems		
Personal Competence <i>Social Competence</i>	As part of the module, concepts learned in the lecture will be implemented on a hardware platform within small groups. Students learn to work in a team and to develop solutions together. Specific tasks are worked on within the group, whereby cross-group collaboration (exchange) also takes place. The second part is a challenge-based project in which the groups find the most energy-efficient solutions possible in healthy competition with each other. This strengthens the cohesion in the groups and reinforces mutual motivation, support and creativity.		
<i>Autonomy</i>	After completing this module, students will be able to independently develop, optimize and evaluate solutions for embedded systems based on the knowledge they have acquired and further technical literature.		
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70		
Credit points	6		
Course achievement	None		
Examination	Written exam		
Examination duration and scale	90 min		
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Nanoelectronics and Microsystems Technology: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Embedded Systems: Elective Compulsory		

Course L2870: Energy Efficiency in Embedded Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Ulf Kulau
Language	EN
Cycle	WiSe
Content	Motivation: In the field of computer science we have only limited possibilities to influence the efficiency of the hardware directly, respectively we are dependent on the manufacturers (e.g. of microcontrollers). However, in order to exploit the full potential of the hardware we are given at the system level, we need a deeper understanding of the background, processes and mechanisms of power dissipation in embedded systems. Where does the power dissipation come from, what happens at the hardware level, what mechanisms can I use directly/indirectly, what is the tradeoff between flexibility and efficiency,..... are only a few questions, which will be elaborated and discussed in this event. Contents of teaching: • Motivation and power dissipation on semiconductor level • Power dissipation of digital circuits, in particular CMOS • Power Management in Hard- and Software (Sleep Modes, DVS, FS, Undervolting) • Energy efficient system design (applications) • Energy Harvesting and Transiently Powered Computing (TPC)
Literature	DE: Die Vorlesungsinhalte wurden in einem Paper veröffentlicht. All Quellen, welche in diesem Kurs verwendet wurden sind in folgenden Paper aufgelistet: ENG: The lecture contents were published in a paper. All sources used in this course are listed in the following paper: 1. Kulau, Ulf: Course: Energy Efficiency in Embedded Systems-A System-Level Perspective for Computer Scientists, EWME, 2018. DE: Als sehr gute Quellen für notwendiges Hintergrundwissen eignen sich die folgenden Bücher: ENG: The following books are very good sources for the necessary background knowledge: 1. Harris, David, and N. Weste: CMOS VLSI Design ed., Pearson Education, 2010 2. Rabaey, Jan: Low Power Design Essentials (Integrated Circuits and Systems), Springer, 2009

Course L2872: Energy Efficiency in Embedded Systems	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Ulf Kulau
Language	EN
Cycle	WiSe
Content	In this project-based exercise, the learned aspects for achieving energy-efficient embedded systems are implemented and consolidated in practical environments in a small project. First, a tool set for the implementation of energy efficiency mechanisms is implemented in common exercises by means of defined tasks. In the second part, a challenge-based exercise is carried out in which a system that is as efficient as possible is to be implemented independently. A system based on an AVR micro-controller is used, which can be operated autonomously by a Solar-Energy Harvester. 1. Task phase: 6 "hands-on" tasks to gain experience and to create a SW library. 2. Project phase: Implementation of an energy autonomous system with the goal of highest possible energy efficiency (Challenge)
Literature	DE: Die Vorlesungsinhalte wurden in einem Paper veröffentlicht. Alle Quellen, welche in diesem Kurs (inklusive Übungen und PBL) verwendet wurden sind in folgenden Paper aufgelistet: ENG: The lecture contents were published in a paper. All sources used in this course (including exercises and PBL) are listed in the following paper: 1. Kulau, Ulf: Course: Energy Efficiency in Embedded Systems-A System-Level Perspective for Computer Scientists, EWME, 2018. DE: Als sehr gute Quellen für notwendiges Hintergrundwissen eignen sich die folgenden Bücher: ENG: The following books are very good sources for the necessary background knowledge: 1. Harris, David, and N. Weste: CMOS VLSI Design ed., Pearson Education, 2010 2. Rabaey, Jan: Low Power Design Essentials (Integrated Circuits and Systems), Springer, 2009

Course L2871: Energy Efficiency in Embedded Systems	
Typ	Recitation Section (large)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Ulf Kulau
Language	EN
Cycle	WiSe
Content	In the lecture hall exercise, the theoretical basics taught in the lecture are deepened. This is done through in-depth discussion of relevant aspects, but also through calculation examples, in which a deeper understanding of the topic of energy efficiency in embedded systems is gained. Exercises will be distributed in advance and solutions will be presented in the lecture hall exercise. Contents of the exercise are as follows: • Basics and calculation of power dissipation on semiconductor • Power dissipation of CMOS using the example of an inverter • Influence of the activity factor and external components • DVS and scheduling • Evaluation to show the benefit of undervolting • Aspects of energy harvesting (MPPT) •
Literature	DE: Die Vorlesungsinhalte wurden in einem Paper veröffentlicht. Alle Quellen, welche in diesem Kurs (also auch den Übungen) verwendet wurden sind in folgenden Paper aufgelistet: ENG: The lecture contents were published in a paper. All sources used in this course (including exercises, and PBL) are listed in the following paper: 1. Kulau, Ulf: Course: Energy Efficiency in Embedded Systems-A System-Level Perspective for Computer Scientists, EWME, 2018. DE: Als sehr gute Quellen für notwendiges Hintergrundwissen eignen sich die folgenden Bücher: ENG: The following books are very good sources for the necessary background knowledge: 1. Harris, David, and N. Weste: CMOS VLSI Design ed., Pearson Education, 2010 2. Rabaey, Jan: Low Power Design Essentials (Integrated Circuits and Systems), Springer, 2009

Module M1700: Satellite Communications and Navigation			
Courses			
Title	Typ	Hrs/wk	CP
Radio-Based Positioning and Navigation (L2711)	Lecture	2	3
Satellite Communications (L2710)	Lecture	3	3
Module Responsible	Prof. Gerhard Bauch		
Admission Requirements	None		
Recommended Previous Knowledge	The module is designed for a diverse audience, i.e. students with different background. Basic knowledge of communications engineering and signal processing are of advantage but not required. The course intends to provide the chapters on communications techniques such that on the one hand students with a communications engineering background learn additional concepts and examples (e.g. modulation and coding schemes or signal processing concepts) which have not or in a different way been treated in our other bachelor and master courses. On the other hand, students with other background shall be able to grasp the ideas but may not be able to understand in the same depth. The individual background of the students will be taken into consideration in the oral exam.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> The students are able to understand, compare and analyse digital satellite communications system as well as navigation techniques. They are familiar with principal ideas of the respective communications, signal processing and positioning methods. They can describe distortions and resulting limitations caused by transmission channels and hardware components. They can describe how fundamental communications and navigation techniques are applied in selected practical systems. The students are familiar with the contents of lecture and tutorials. They can explain and apply them to new problems. <i>Skills</i> The students are able to describe and analyse digital satellite communications systems and navigation systems. They are able to analyse transmission chains including link budget calculations. They are able to choose appropriate transmission technologies and system parameters for given scenarios. Personal Competence <i>Social Competence</i> The students can jointly solve specific problems. <i>Autonomy</i> The students are able to acquire relevant information from appropriate literature sources.		
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70		
Credit points	6		
Course achievement	None		
Examination	Oral exam		
Examination duration and scale	30 min		
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory		

Course L2711: Radio-Based Positioning and Navigation	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Gerhard Bauch, Dr. Rico Mendrzik
Language	EN
Cycle	SoSe
Content	- Information extraction from communication signals - Time-of-arrival principle - Ranging in additive white Gaussian noise (AWGN) channel - Correlation-based range estimation - Effect of multipath propagation on time-of-arrival principle - Zero-forcing range estimation in the presence of multipath - Optimum range estimation in the presence of multipath - Zero-forcing in presence of noise - Angle-of-arrival principle - Angle-of-arrival estimation in AWGN channel - Delay-and-sum estimator - Multiple Signal Classifier (MUSIC) - MUSIC-based angle-of-arrival estimation - Case study: Comparison of estimators in AWGN channels - Effect of multipath propagation on angle-of-arrival principle - Case study: Comparison of estimators in multipath channels

- Information fusion of extracted signals
 - Distance-based positioning
 - Principle of time-of-arrival positioning
 - Geometric interpretation
 - Positioning in the absence of noise
 - Linearization of the positioning problem
 - Positioning in the presence of noise
 - Optimality criteria
 - Least squares time-of-arrival positioning
 - Maximum likelihood time-of-arrival positioning
 - Interactive Matlab demo
 - Excursion: gradient descent solvers for nonlinear programs
 - Real-life positioning with embedded development board (Arduino)
 - Linearized least squares time-of-arrival positioning
 - Effect of clock offsets on distance-based positioning
 - Time-difference-of-arrival principle
 - Least squares time-difference-of-arrival positioning
 - Clock offset mitigation via two-way ranging
 - Performance limits of distance-based positioning
 - Fisher information and the Cramér-Rao lower bound
 - Fisher information in the AWGN case
 - Multi-variate Fisher information
 - Cramér-Rao lower bound for synchronized time-of-arrival positioning
 - Case study: Synchronized time-of-arrival positioning
 - Cramér-Rao lower bound for unsynchronized time-of-arrival positioning
 - Case study: Unsynchronized time-of-arrival positioning
 - Angle-based Positioning
 - Angle-of-arrival positioning principle
 - Geometric interpretation angle-of-arrival positioning principle
 - Noise-free angle-of-arrival positioning with known orientation
 - Effect of noise on angle-of-arrival positioning
 - Least squares angle-of-arrival positioning with known orientation
 - Linear least squares angle-of-arrival positioning
 - Effect of orientation uncertainty
 - Angle-difference-of-arrival positioning
 - Geometric interpretation angle difference of arrival positioning
 - Proof of angle-difference-of-arrival locus
 - Inscribed angle lemma
 - Case study: Angle-difference-of-arrival-positioning
 - Performance limits of angle-based positioning
 - Cramér-Rao lower bound for angle-of-arrival positioning with known orientation
 - Case study: Angle-of-arrival positioning with known orientation
- Information Filtering
 - Bayesian filtering
 - Principle of Bayesian filtering
 - General Problem Formulation
 - Solution to the linear Gaussian case
 - State transition in the linear Gaussian case
 - Proof of predicted posterior distribution of the Kalman filter
 - State update in the linear Gaussian case
 - Proof of marginal posterior distribution of the Kalman filter
 - Working with Gaussian random variables
 - Proof: Affine transformation
 - Proof: Marginalization
 - Proof: Conditioning
 - Kalman filter: Optimum Inference in the linear Gaussian case
 - Modeling of process noise
 - Modeling of measurement noise
 - Case study: Kalman filtering in the linear Gaussian case
 - Interactive Kalman filtering in Matlab
 - Dealing with nonlinearities in Bayesian filtering
 - Nonlinear Gaussian case
 - Extended Kalman filter
 - Proof of predicted posterior distribution of the extended Kalman filter
 - Proof of marginal posterior distribution of the extended Kalman filter
 - Example: Nonlinear state transition
 - Case study: Extended Kalman filtering
 - Practical considerations for filter design
- Satellite Navigation
 - Overview from positioning perspective
 - Earth-centered earth-fixed (ECEF) coordinate system
 - World geodetic system (WGS)
 - Satellite navigation systems
 - System-receiver clock offsets and pseudo-ranges

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	▪ Unsynchronized time-of-arrival positioning revisited ◦ GPS legacy signals and ranging ▪ Signal overview ▪ Time-of-arrival principle revisited ▪ Direct sequence spread spectrum principle ▪ Short and long codes ▪ Satellite signal generation ▪ Carriers and codes ▪ Correlation properties of codes ▪ Code division multiple access in flat fading channels ▪ Navigation message ◦ Velocity estimation ◦ Hands-on case study: Design of an extended Kalman filter for satellite navigation based on recorded data • Robust navigation ◦ Multipath-assisted positioning in millimeter wave multiple antenna systems ◦ Multi-sensor fusion
Literature	

Course L2710: Satellite Communications	
Typ	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	SoSe
Content	• Introduction to satellite communications ◦ What is a satellite ◦ Overview orbits, Van Allen Belt, components of a satellite ◦ Satellite services ◦ Frequency bands for satellite services ◦ International Telecommunications Union (ITU) ◦ Influence of atmospheric impairments ◦ Milestones in satellite communications • Components of a satellite communications system ◦ Ground segment ◦ Space segment ◦ Control segment • Communication links ◦ Uplink, downlink ◦ Forward link, reverse link ◦ Intersatellite links ◦ Multiple access ◦ Performance measures ▪ Effective isotropic radiated power (EIRP), antenna gain, figure of merit, G/T, carrier to noise ratio ▪ Signal to noise power ratio vs. carrier to noise ratio • Single beam and multibeam satellites ◦ Beam coverage ◦ Examples for beam coverage of LEO and GEO satellites (Iridium, Viasat) • Transparent vs. regenerative payload • Orbits ◦ Low earth orbit (LEO), medium earth orbit (MEO), geosynchronous and geostationary orbits (GEO), highly elliptical orbits (HEO) ◦ Favourable orbits: ▪ HEO orbits with 63-64° inclination, Molnya and Tundra orbits ▪ Circular LEO orbits ▪ Circular MEO Orbits (Intermediate Circular Orbits (ICO)) ▪ Equatorial orbits, geostationary orbit (GEO) ◦ Important aspects of LEO, MEO and GEO satellites • Kepler's laws of planetary motion • Gravitational force • Parameters of ellipses and elliptical orbits ◦ Major and minor half axis ◦ Foci ◦ Eccentricity ◦ Eccentric anomaly, mean anomaly, true anomaly ◦ Area ◦ Orbit period ◦ Perigee, apogee ◦ Distance of satellite from center of earth ◦ Construction of ellipses according to de La Hire ◦ Orbital plane in space, inclination, right ascension (longitude) of ascending node, Vernal equinox

- Newton's laws of motion
- Newton's universal law of gravitation
- Energy of satellites: Potential energy, kinetic energy, total energy
- Instantaneous speed of a satellite
- Kepler's equation
- Satellite visibility, elevation
- Required number of LEO, MEO or GEO satellites for continuous earth coverage
- Satellite altitude and distance from a point on earth
- Choice of orbits
 - LEO, HEO, GEO
 - Elliptical orbits with non-zero inclination, Molnya orbits, Tundra orbits
 - Geosynchronous orbits
 - Parameters of geosynchronous orbits
 - Circular geosynchronous orbits
 - Inclined geosynchronous orbits
 - Quasi-zenith satellite systems (QZSS)
 - Syb-synchronous circular equatorial orbits
 - Geostationary orbit
 - Parameters of the geostationary orbit
 - Visibility
 - Propagation delay
 - Applications and system examples
- Perturbations of orbits
 - Station keeping
 - Station keeping box
 - Estimation of orbit parameters
- Fundamentals of digital communications techniques
 - Components of a digital communications system
 - Principles of encryption
 - Scrambling
 - Scrambling vs. interleaving for randomization of data sequences
 - Interleaving: Block interleaver, convolutional interleaver, random interleaver
 - Digital modulation methods
 - Linear and non-linear digital modulation methods
 - Linear digital modulation methods
 - QAM modulator and demodulator
 - Pulse shaping, square-root raised-cosine pulses
 - Average power spectral density
 - Signal space constellation
 - Examples: M-ary phase shift keying (M-PSK), M-ary quadrature amplitude shift keying (M-QAM)
 - M-PSK in noisy channels
 - Bit error probabilities of M-PSK and M-QAM
 - M-PSK vs. M-QAM
 - M-ary amplitude and phase shift keying (M-APSK)
 - M-APSK vs. M-QAM
 - Differential phase shift keying (DPSK)

Error control coding (channel coding)

- Error detecting and forward error correcting (FEC) codes
- Principle of channel coding
- Data rate, code rate, Baud rate, spectral efficiency of modulation and coding schemes
- Bandwidth-power trade-off, bandwidth-limited vs. power-limited transmission
- Coding and modulation for transparent vs. regenerative payload
- Block codes and convolutional codes
- Concatenated codes
- Bit-interleaved coded modulation
- Convolutional codes
- Low density parity check (LDPC) codes, principle of message passing decoding, bit error rate performance
- Cyclic block codes
 - Examples for cyclic block codes
 - Single errors vs. block errors, cyclic block codes for burst errors
 - Generator matrix, generator polynomials
 - Systematic encoding and syndrome determination with shift registers
 - Cyclic redundancy check (CRC) codes
- Automatic repeat request (ARQ)
 - Principle of ARQ
 - Stop-and-wait ARQ
 - Go-back-N ARQ
 - Selective-repeat ARQ
- Transmission gains and losses

	◦ Antenna gain ■ Antenna radiation pattern ■ Maximum antenna gain, 3dB beamwidth ■ Maximum antenna gain of circular aperture ■ Maximum antenna gain of a geostationary satellite with global coverage ◦ Effective isotropic radiated power (EIRP) ◦ Power flux density ◦ Path loss ■ Free space loss, free space loss for geostationary satellites ■ Atmospheric loss ■ Received power ◦ Losses in transmit and receive equipment ■ Feeder loss ■ Depointing loss ■ Polarization mismatch loss ◦ Combined effect of losses • Noise ◦ Origins of noise ◦ White noise ◦ Noise power spectral density and noise power ◦ Additive white Gaussian noise (AWGN) channel model ◦ Antenna noise temperature ◦ Earth brightness temperature ◦ Signal to noise ratios • Atmospheric distortions ◦ Atmosphere of the earth: Troposphere, stratosphere, mesosphere, thermosphere, exosphere ◦ Attenuation and depolarization due to rain, fog, rain and ice clouds, sandstorms ◦ Scintillation ◦ Faraday effect ◦ Multipath contributions • Link budget calculations ◦ GEO clear sky uplink and downlink ◦ GEO uplink and downlink under rain conditions ◦ Transparent vs. regenerative payload • Link availability improvement through site diversity and adaptive transmission ◦ Transparent vs. regenerative payload ■ Non-linear amplifiers ■ Saleh model, Rapp model ■ Input and output back-off factor ■ Single carrier and multicarrier operation ■ Dimensioning of transmission parameters ■ Sources of noise: Thermal noise, interference, intermodulation products ■ Signal to noise ratio and bit error probability ■ Robustness against interference and non-linear channels • Satellite networks ◦ Satellite network reference architectures ◦ Network topologies ◦ Network connectivity ■ Types of network connectivity ■ On-board connectivity ■ Inter-satellite links ◦ Broadcast networks ◦ Satellite-based internet • Satellite communications systems and standards examples ◦ The role of standards in satellite communications ◦ The Digital Video Broadcast Satellite Standard: DVB-S, DVB-S2, DVB-S2X ◦ Satellites in 3GPP mobile communications networks ◦ LEO megaconstellations: SpaceX Starlink, Kuiper, OneWeb ◦ Space debris ◦ IRIS² (Infrastructure for Resilience, Interconnectivity and Security by Satellite): The new EU Secure Satellite Constellation ◦ The German Heinrich Hertz mission
Literature	

Module M1871: Big Data				
Courses				
Title	Typ		Hrs/wk	CP
Big Data (L3101)	Lecture		2	3
Big Data (L3102)	Project-/problem-based Learning		2	3
Module Responsible	Prof. Stefan Schulte			
Admission Requirements	None			
Recommended Previous Knowledge	• Good software engineering and implementation skills are very important for the practical part of this module • Previous knowledge in the fields of Distributed Systems and Databases is helpful			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> After successful completion of the course, students are able to: • Describe basic concepts of methods and technologies to process and store very large amounts of data • Discuss and assess critical aspects of Big Data • Select and apply Big Data technologies for particular application areas • Design and develop practical solutions for the processing of very large amounts of data • Implement Big Data services <i>Skills</i> The students acquire the ability to model Big Data systems and to work with these systems. This comprises especially the ability to select and utilize fitting technologies for different application areas. Furthermore, students are able to critically assess the chosen technologies. Personal Competence <i>Social Competence</i> Students can work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve the problem. <i>Autonomy</i> Students are able to independently investigate a complex problem and assess which competencies are required to solve it.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Subject theoretical and practical work			
Examination duration and scale	Group project incl. presentation, written exam			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Core Qualification: Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory			

Course L3101: Big Data	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Stefan Schulte
Language	EN
Cycle	SoSe
Content	This lecture discusses the fundamental concepts of modern Big Data systems. The following topics will be covered in the single lectures: • Cloud Computing • Data models for Big Data • NoSQL databases • Lambda architecture including batch, speed, and serving layers • Kappa architecture • Concepts and tools for data stream processing • Current software tools for the processing of very large amounts of data
Literature	Lecture notes as well as current literature, are announced in the lecture.

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Course L3102: Big Data	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Stefan Schulte
Language	EN
Cycle	SoSe
Content	This project-/problem-oriented part of the module augments the theoretical content of the lecture with a concrete technical problem, which needs to be solved by the students in group work during the semester. The topics are from the field of Big Data, and address for instance data stream processing or batch processing.
Literature	Lecture notes as well as current literature announced in the lecture.

Module M2029: Process Imaging				
Courses				
Title	Typ		Hrs/wk	CP
Process Imaging (L2723)	Lecture		3	3
Applied Process Imaging (L2724)	Project-/problem-based Learning		3	3
Module Responsible	Prof. Alexander Penn			
Admission Requirements	None			
Recommended Previous Knowledge	No special prerequisites needed. An interest in imaging techniques and image processing is helpful but not mandatory.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	The module focuses primarily on discussing established imaging techniques including (a) optical and infrared imaging, (b) magnetic resonance imaging, (c) X-ray imaging and tomography. Moreover, it presents and discusses a range of more recent imaging modalities. The students will learn: 1. what these imaging techniques can measure (such as sample density or concentration, material transport, chemical composition, temperature), 2. how the measurement techniques work (physical measurement principles, hardware requirements, image reconstruction), and 3. how to determine the most suited imaging methods for a given problem. After the successful completion of the course, the students shall: 1. understand the physical principles and practical aspects of the most common imaging methods, 2. be able to assess the pros and cons of these methods with regard to cost, complexity, expected contrasts, spatial and temporal resolution, and based on this assessment 3. be able to identify the most suited imaging modality for any specific engineering challenge in the field of chemical and bioprocess engineering.			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>	In the problem-based interactive course, students work in small teams and set up two process imaging systems and use these systems to measure relevant process parameters in different chemical and bioprocess engineering applications. The teamwork will foster interpersonal communication skills.			
<i>Autonomy</i>	Students are guided to work in self-motivation due to the challenge-based character of this module. A final presentation improves presentation skills.			
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84			
Credit points	6			
Course achievement	None			
Examination	Subject theoretical and practical work			
Examination duration and scale	70% written examination, 30% active participation and final presentation of the problem-based learning units with a 5-10 page report			
Assignment for the Following Curricula	Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Bioprocess Engineering: Specialisation B - Industrial Bioprocess Engineering: Elective Compulsory Bioprocess Engineering: Specialisation C - Bioeconomic Process Engineering, Focus Energy and Bioprocess Technology: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Specialisation Chemical and Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Core Qualification: Elective Compulsory Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory International Management and Engineering: Specialisation II. Process Engineering and Biotechnology: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory Process Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Process Engineering: Specialisation Environmental Process Engineering: Elective Compulsory			

Course L2723: Process Imaging	
Typ	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Alexander Penn
Language	EN
Cycle	SoSe
Content	The lecture focuses primarily on presenting and discussing established imaging techniques relevant to the field of engineering including (a) optical and infrared imaging, (b) magnetic resonance imaging, (c) X-ray imaging and tomography. Moreover, it presents and discusses a range of more recent imaging modalities. The students will learn: 1. what these imaging techniques can measure (such as sample density or concentration, material transport, chemical composition, temperature), 2. how the measurement techniques work (physical measurement principles, hardware requirements, image reconstruction), and 3. how to determine the most suited imaging methods for a given problem.
Literature	Wang, M. (2015). Industrial Tomography. Cambridge, UK: Woodhead Publishing. Available as e-book in the library of TUHH: https://katalog.tub.tuhh.de/Record/823579395

Course L2724: Applied Process Imaging	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Alexander Penn, Dr. Stefan Benders
Language	EN
Cycle	SoSe
Content	Content: The module focuses primarily on discussing established imaging techniques including (a) optical and infrared imaging, (b) magnetic resonance imaging, (c) X-ray imaging and tomography, and (d) ultrasound imaging and also covers a range of more recent imaging modalities. The students will learn: 1. what these imaging techniques can measure (such as sample density or concentration, material transport, chemical composition, temperature), 2. how the measurements work (physical measurement principles, hardware requirements, image reconstruction), and 3. how to determine the most suited imaging methods for a given problem. Learning goals: After the successful completion of the course, the students shall: 1. understand the physical principles and practical aspects of the most common imaging methods, 2. be able to assess the pros and cons of these methods with regard to cost, complexity, expected contrasts, spatial and temporal resolution, and based on this assessment 3. be able to identify the most suited imaging modality for any specific engineering challenge in the field of chemical and bioprocess engineering.
Literature	Wang, M. (2015). Industrial Tomography. Cambridge, UK: Woodhead Publishing. Available as e-book in the library of TUHH: https://katalog.tub.tuhh.de/Record/823579395

Module M1598: Image Processing				
Courses				
Title	Typ		Hrs/wk	CP
Image Processing (L2443)	Lecture		2	4
Image Processing (L2444)	Recitation Section (small)		2	2
Module Responsible	Prof. Tobias Knopp			
Admission Requirements	None			
Recommended Previous Knowledge	Signal and Systems			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students know about • visual perception • multidimensional signal processing • sampling and sampling theorem • filtering • image enhancement • edge detection • multi-resolution procedures: Gauss and Laplace pyramid, wavelets • image compression • image segmentation • morphological image processing <i>Skills</i> The students can • analyze, process, and improve multidimensional image data • implement simple compression algorithms • design custom filters for specific applications Personal Competence <i>Social Competence</i> Students can work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve the problem. <i>Autonomy</i> Students are able to independently investigate a complex problem and assess which competencies are required to solve it.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Data Science: Specialisation I. Mathematics/Computer Science: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Medical Technology: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L2443: Image Processing	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Tobias Knopp
Language	EN
Cycle	WiSe
Content	• Visual perception • Multidimensional signal processing • Sampling and sampling theorem • Filtering • Image enhancement • Edge detection • Multi-resolution procedures: Gauss and Laplace pyramid, wavelets • Image Compression • Segmentation • Morphological image processing
Literature	Bredies/Lorenz, Mathematische Bildverarbeitung, Vieweg, 2011 Pratt, Digital Image Processing, Wiley, 2001 Bernd Jähne: Digitale Bildverarbeitung - Springer, Berlin 2005

Course L2444: Image Processing	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Tobias Knopp
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Focus Software

Module M0753: Software Verification

Courses

Title	Typ	Hrs/wk	CP
Software Verification (L0629)	Lecture	2	3
Software Verification (L0630)	Recitation Section (small)	2	3

Module Responsible	Prof. Sibylle Schupp
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Admission Requirements	None
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Recommended Previous Knowledge	- Automata theory and formal languages - Computational logic - Object-oriented programming, algorithms, and data structures - Functional programming or procedural programming - Concurrency
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Educational Objectives	After taking part successfully, students have reached the following learning results
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Professional Competence	
<i>Knowledge</i>	Students apply the major verification techniques in model checking and deductive verification. They explain in formal terms syntax and semantics of the underlying logics, and assess the expressivity of different logics as well as their limitations. They classify formal properties of software systems. They find flaws in formal arguments, arising from modeling artifacts or underspecification.
<i>Skills</i>	Students formulate provable properties of a software system in a formal language. They develop logic-based models that properly abstract from the software under verification and, where necessary, adapt model or property. They construct proofs and property checks by hand or using tools for model checking or deductive verification, and reflect on the scope of the results. Presented with a verification problem in natural language, they select the appropriate verification technique and justify their choice.
Personal Competence	
<i>Social Competence</i>	Students discuss relevant topics in class. They defend their solutions orally. They communicate in English.
<i>Autonomy</i>	Using accompanying on-line material for self study, students can assess their level of knowledge continuously and adjust it appropriately. Working on exercise problems, they receive additional feedback. Within limits, they can set their own learning goals. Upon successful completion, students can identify and precisely formulate new problems in academic or applied research in the field of software verification. Within this field, they can conduct independent studies to acquire the necessary competencies and compile their findings in academic reports. They can devise plans to arrive at new solutions or assess existing ones.

Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
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Credit points	6
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Course achievement	Compulsory	Bonus	Form	Description
	Yes	15 %	Exercises	

Examination	Written exam
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Examination duration and scale	90 min
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Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory
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Course L0629: Software Verification

Typ	Lecture
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Hrs/wk	2
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CP	3
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Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
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Lecturer	Prof. Sibylle Schupp
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Language	EN
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Cycle	WiSe
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Content	- Model checking (bounded model checking, CTL, LTL) - Real-time model checking (TCTL, timed automata) - Deductive verification (Hoare logic) - Tool support - Recent developments of verification techniques and applications
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Literature	C. Baier and J-P. Katoen, Principles of Model Checking, MIT Press 2007. M. Huth and M. Bryan, Logic in Computer Science. Modelling and Reasoning about Systems, 2nd Edition, 2004. - Selected Research Papers
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Course L0630: Software Verification	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1397: Model Checking - Proof Engines and Algorithms				
Courses				
Title	Typ		Hrs/wk	CP
Model Checking - Proof Engines and Algorithms (L1979)	Lecture		2	3
Model Checking - Proof Engines and Algorithms (L1980)	Recitation Section (small)		2	3
Module Responsible	Prof. Görschwin Fey			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge about data structures and algorithms			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students know - algorithms and data structures for model checking, - basics of Boolean reasoning engines and - the impact of specification and modelling on the computational effort for model checking. <i>Skills</i> Students can - explain and implement algorithms and data structures for model checking, - decide whether a given problem can be solved using Boolean reasoning or model checking, and - implement the respective algorithms. Personal Competence <i>Social Competence</i> Students - discuss relevant topics in class and - defend their solutions orally. <i>Autonomy</i> Using accompanying material students independently learn in-depth relations between concepts explained in the lecture and additional solution strategies.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Subject theoretical practical work	andDie Aufgabe wird im Rahmen von Volresung und Prüfung definiert. Die Lösung der Aufgabe ist Zulassungsvoraussetzung für die Prüfung.
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory			

Course L1979: Model Checking - Proof Engines and Algorithms	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Görschwin Fey
Language	EN
Cycle	SoSe
Content	Correctness is a major concern in embedded systems. Model checking can fully automatically proof formal properties about digital hardware or software. Such properties are given in temporal logic, e.g., to prove "No two orthogonal traffic lights will ever be green." And how do the underlying reasoning algorithms work so effectively in practice despite a computational complexity of NP hardness and beyond? But what are the limitations of model checking? How are the models generated from a given design? The lecture will answer these questions. Open source tools will be used to gather a practical experience. Among other topics, the lecture will consider the following topics: • Modelling digital Hardware, Software, and Cyber Physical Systems • Data structures, decision procedures and proof engines ◦ Binary Decision Diagrams ◦ And-Inverter-Graphs ◦ Boolean Satisfiability ◦ Satisfiability Modulo Theories • Specification Languages ◦ CTL ◦ LTL ◦ System Verilog Assertions • Algorithms for ◦ Reachability Analysis ◦ Symbolic CTL Checking ◦ Bounded LTL-Model Checking ◦ Optimizations, e.g., induction, abstraction • Quality assurance
Literature	Edmund M. Clarke, Jr., Orna Grumberg, and Doron A. Peled. 1999. <i>Model Checking</i>. MIT Press, Cambridge, MA, USA. A. Biere, A. Biere, M. Heule, H. van Maaren, and T. Walsh. 2009. <i>Handbook of Satisfiability: Volume 185 Frontiers in Artificial Intelligence and Applications</i>. IOS Press, Amsterdam, The Netherlands, The Netherlands. Selected research papers

Course L1980: Model Checking - Proof Engines and Algorithms	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Görschwin Fey
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1301: Software Testing				
Courses				
Title	Typ		Hrs/wk	CP
Software Testing (L1791)	Lecture		2	3
Software Testing (L1792)	Project-/problem-based Learning		2	3
Module Responsible	Prof. Sibylle Schupp			
Admission Requirements	None			
Recommended Previous Knowledge	• Software Engineering • Higher Programming Languages • Object-Oriented Programming • Algorithms and Data Structures • Experience with (Small) Software Projects • Statistics			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students explain the different phases of testing, describe fundamental techniques of different types of testing, and paraphrase the basic principles of the corresponding test process. They give examples of software development scenarios and the corresponding test type and technique. They explain algorithms used for particular testing techniques and describe possible advantages and limitations. <i>Skills</i> Students identify the appropriate testing type and technique for a given problem. They adapt and execute respective algorithms to execute a concrete test technique properly. They interpret testing results and execute corresponding steps for proper re-test scenarios. They write and analyze test specifications. They apply bug finding techniques for non-trivial problems. Personal Competence <i>Social Competence</i> Students discuss relevant topics in class. They defend their solutions orally. They communicate in English. <i>Autonomy</i> Students can assess their level of knowledge continuously and adjust it appropriately, based on feedback and on self-guided studies. Within limits, they can : own learning goals. Upon successful completion, students can identify and precisely formulate new problems in academic or applied research in the field of : testing. Within this field, they can conduct independent studies to acquire the necessary competencies and compile their findings in academic reports. T devise plans to arrive at new solutions or assess existing ones			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Subject theoretical and practical work			
Examination duration and scale	Software			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory			

Course L1791: Software Testing	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	SoSe
Content	• Fundamentals of software testing • Model-based testing • Test automation • Criteria-based testing
Literature	• P. Ammann and J. Offutt, "Introduction to Software Testing", 2nd edition 2016. • Selected research papers

Course L1792: Software Testing	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	SoSe
Content	• Fundamentals of software testing • Model-based testing • Test automation • Criteria-based testing
Literature	• M. Pezze and M. Young, Software Testing and Analysis, John Wiley 2008. • P. Ammann and J. Offutt, "Introduction to Software Testing", 2nd edition 2015.

Module M1682: Secure Software Engineering				
Courses				
Title	Typ		Hrs/wk	CP
Secure Software Engineering (L2667)	Lecture		2	3
Secure Software Engineering (L2668)	Project-/problem-based Learning		2	3
Module Responsible	Prof. Riccardo Scandariato			
Admission Requirements	None			
Recommended Previous Knowledge	Familiarity with basic software engineering concepts (e.g., requirements, design) and basic security concepts (e.g., confidentiality, integrity, availability)			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	Students can: - Elicit security requirements in a software project - Model and document security measures in a software design - Use threat and risk analysis techniques - Understand how security code reviews are performed - Understand the core definitions of concepts related to privacy - Understand privacy enhancing technologies			
<i>Skills</i>	Select appropriate security assurance techniques to be used in a security assurance program			
Personal Competence <i>Social Competence</i>	None			
<i>Autonomy</i>	Students can apply the knowledge acquired throughout the course to the resolution of industrial case studies. Students should also be capable to acquire new knowledge independently from academic publications, technical standards, and white papers.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	5 %	Subject theoretical and practical work	and Gruppenarbeit mit aktuellen Technologien aus dem Bereich Sicherheit
Examination	Written exam			
Examination duration and scale	120 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory			

Course L2667: Secure Software Engineering	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	SoSe
Content	- Secure software development processes and maturity models - Techniques to define security requirements - Techniques to create, document and analyse the design of secure applications - Threat and risk analysis techniques - Security code reviews - Program repair techniques for security vulnerabilities - Privacy engineering
Literature	Charles Haley, Robin Laney, Jonathan Moffett, Bashar Nuseibeh, 2008, Security Requirements Engineering: A Framework for Representation and Analysis, Transactions on Software Engineering Sindre, G. and Opdahl, A.L., 2005. Eliciting security requirements with misuse cases. Requirements engineering, 10(1) Fontaine, P.J., Van Lamsweerde, A., Letier, E. and Darimont, R., 2001. Goal-oriented elaboration of security requirements. Joanna C. S. Santos, Katy Tarrit, Mehdi Mirakhorli, 2017, A Catalog of Security Architecture Weaknesses, IEEE International Conference on Software Architecture Workshops (ICSAW) Jürjens, J., 2002, UMLsec: Extending UML for secure systems development, International Conference on The Unified Modeling Language Lund, M.S., Solhaug, B. and Stølen, K., 2011. A guided tour of the CORAS method. In Model-Driven Risk Analysis, Springer Pratyusa Manadhata, Jeannette Wing, 2021, An Attack Surface Metric, IEEE Transactions on Software Engineering, 37(3) Howard, M.A., 2006. A process for performing security code reviews. IEEE Security & privacy, 4(4) Jaap-Henk Hoepman, 2022, Privacy Design Strategies (The Little Blue Book) Diaz, C. and Gürses, S., 2012. Understanding the landscape of privacy technologies. Proceedings of the information security summit, 12

Course L2668: Secure Software Engineering	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1774: Advanced Internet Computing			
Courses			
Title	Typ	Hrs/wk	CP
Advanced Internet Computing (L2916)	Lecture	2	3
Advanced Internet Computing (L2917)	Project-/problem-based Learning	2	3
Module Responsible	Prof. Stefan Schulte		
Admission Requirements	None		
Recommended Previous Knowledge	Good programming skills are necessary. Previous knowledge in the field of distributed systems is helpful.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	After successful completion of the course, students are able to: - Describe basic concepts of cloud computing and blockchain technologies - Discuss and assess critical aspects of cloud computing and blockchain technologies - Select and apply cloud and blockchain technologies for particular application areas - Design and develop practical solutions based on cloud and blockchain software The students acquire the ability to model Internet-based distributed systems and to work with these systems. This comprises especially the ability to select and utilize fitting technologies for different application areas. Furthermore, students are able to critically assess the chosen technologies. Students can work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve the problem. Students are able to independently investigate a complex problem and assess which competencies are required to solve it.		
<i>Knowledge</i>			
<i>Skills</i>			
Personal Competence			
<i>Social Competence</i>			
<i>Autonomy</i>			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Subject theoretical and practical work		
Examination duration and scale	Group project incl. presentation (50 %), written exam (60 min, 50 %)		
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory		

Course L2916: Advanced Internet Computing	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Stefan Schulte
Language	EN
Cycle	SoSe
Content	This lecture discusses modern Internet-based distributed systems in two blocks: (i) cloud computing and (ii) blockchain technologies. The following topics will be covered in the single lectures: - Cloud Service and Deployment Models - Virtualization - Further Aspects of Cloud Computing - Edge and Fog Computing - Blockchain technologies - Consensus
Literature	Lecture notes as well as current literature announced in the lecture.

Module Manual M.Sc. "Information and Communication Systems"

Course L2917: Advanced Internet Computing	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Stefan Schulte
Language	EN
Cycle	SoSe
Content	This project-/problem-oriented part of the module augments the theoretical content of the lecture by a concrete technical problem, which needs to be solved by the students in group work during the semester. Possible topics are (blockchain-based) sensor data integration and blockchain-based voting.
Literature	Lecture notes as well as current literature announced in the lecture.

Module M0924: Software for Embedded Systems				
Courses				
Title	Typ		Hrs/wk	CP
Software for Embedded Systems (L1069)	Lecture		2	3
Software for Embedded Systems (L1070)	Project/problem-based Learning		3	3
Module Responsible	Prof. Bernd-Christian Renner			
Admission Requirements	None			
Recommended Previous Knowledge	• Very good knowledge and practical experience in programming in the C language and its compilation process • Basic knowledge in software engineering • Basic understanding of assembly language • Basic knowledge of electrical engineering			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Knowledge • Students know the contents of the module and can reproduce and link them. • They are able to describe the usage and advantages of event-based programming using interrupts. • Students can explain the basic principles and procedures for creating software for embedded systems. • They know the components and functions of a concrete microcontroller. • The participants can explain requirements of real time systems. • They know at least three scheduling algorithms for real time operating systems including their pros and cons. • They know different memories for embedded systems and their use cases, including advantages and disadvantages. • They understand race conditions, their causes and consequences. • They are familiar with at least two bootloader concepts and can evaluate their suitability in different contexts. • They know mechanisms for optimizing the power consumption of microcontrollers with sleep modes. Skills • Students design and write hardware-oriented software modules in a high-level language (C or similar) for an embedded system based on a specific microcontroller. • They can configure and interact with peripherals (timer, ADC, EEPROM), including bit-level register operations, interrupt-based processing, and program flow. • They implement and use a (preemptive) scheduler for an embedded system. • They learn to write independent, reusable software components. They build a complex software system (library and applications) throughout the semester. • They learn how to model event-based software for embedded systems with UML state diagrams and finite state machines and implement a complex system that they modeled. • They can identify race conditions in unknown software and develop strategies to work around them. Personal Competence Social Competence • Students are able to work goal-oriented in small mixed groups. • They learn and broaden their teamwork abilities. • They learn to define and split tasks within the team. • They practice to present their software, the selection of algorithms and their design choices. Autonomy Students are able • to solve assignments related to this lecture independently with instructional direction. • to design, implement, and test software components for an embedded system independently based on a textual description. • to read and understand data sheets and manuals of electronic components (such as micro-controllers and sensors). • to independently transfer the concepts known from the lecture to the exercises and to select algorithms based on given criteria.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Attestation	
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Embedded Systems: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L1069: Software for Embedded Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	• General-Purpose Processors • Programming the Atmel AVR • Interrupts and Race Conditions • C for Embedded Systems • Standard Single Purpose Processors: Peripherals • Finite-State Machines • Memory • Operating Systems for Embedded Systems • Real-Time Embedded Systems • Boot loader and Power Management
Literature	1. Embedded System Design, F. Vahid and T. Givargis, John Wiley 2. Programming Embedded Systems: With C and Gnu Development Tools, M. Barr and A. Massa, O'Reilly 3. C und C++ für Embedded Systems, F. Bollow, M. Homann, K. Köhn, MITP 4. The Art of Designing Embedded Systems, J. Ganssle, Newnes 5. Mikrocomputertechnik mit Controllern der Atmel AVR-RISC-Familie, G. Schmitt, Oldenbourg 6. Making Embedded Systems: Design Patterns for Great Software, E. White, O'Reilly

Course L1070: Software for Embedded Systems	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1785: Machine Learning in Electrical Engineering and Information Technology			
Courses			
Title	Typ	Hrs/wk	CP
General Introduction Machine Learning (L3004)	Lecture	1	2
Machine Learning Applications in Electric Power Systems (L3008)	Lecture	1	1
Machine Learning in Electromagnetic Compatibility (EMC) Engineering (L3006)	Lecture	1	1
Machine Learning in High-Frequency Technology and Radar (L3007)	Lecture	1	1
Machine Learning in Wireless Communications (L3005)	Lecture	1	1
Module Responsible	Prof. Gerhard Bauch		
Admission Requirements	None		
Recommended Previous Knowledge	The module is designed for a diverse audience, i.e. students with different background. It shall be suitable for both students with deeper knowledge in machine learning methods but less knowledge in electrical engineering, e.g. math or computer science students, and students with deeper knowledge in electrical engineering but less knowledge in machine learning methods, e.g. electrical engineering students. Machine learning methods will be explained on a relatively high level indicating mainly principle ideas. The focus is on specific applications in electrical engineering and information technology. The chapters of the course will be understandable in different depth depending on the individual background of the student. The individual background of the students will be taken into consideration in the oral exam.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> The students know basic machine learning concepts and learning strategies. They are aware of specific opportunities, challenges and approaches of machine learning in various fields of electrical engineering. They know exemplary applications of machine learning in electrical engineering. The students are familiar with the contents of the module courses. They can explain and apply them to new problems. <i>Skills</i> The students are able to apply methods from machine learning to problems in electrical engineering. They are able to determine, dimension and implement suitable approaches such as types of deep learning networks and learning strategies for specific engineering problems. In particular, they are able to include domain knowledge in machine learning architectures and learning strategies. They are able to critically assess the learning results based on domain knowledge. Personal Competence <i>Social Competence</i> The students can jointly solve specific problems. <i>Autonomy</i> The students are able to acquire relevant information from appropriate literature sources. They can control their level of knowledge during the lecture period e.g. by solving tutorial problems or using software tools.		
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70		
Credit points	6		
Course achievement	None		
Examination	Oral exam		
Examination duration and scale	30 min		
Assignment for the Following Curricula	Data Science: Specialisation III. Applications: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Microwave Engineering, Optics, and Electromagnetic Compatibility: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Microwave Engineering, Optics, and Electromagnetic Compatibility: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory		

Module Manual M.Sc. "Information and Communication Systems"

Course L3004: General Introduction Machine Learning	
Typ	Lecture
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Dr. Maximilian Stark
Language	EN
Cycle	SoSe
Content	• From Rule-Based Systems to Machine Learning ◦ Brief overview recent advances in ML in various domain ◦ Outline and expected learning outcomes ◦ Basics statistical inference and statistics ◦ Basics of information theory • The Notions of Learning in Machine Learning ◦ Unsupervised and supervised machine learning ◦ Model-based and data-driven machine learning ◦ Hybrid modelling ◦ Online/offline/meta/transfer learning ◦ General loss functions • Introduction to Deep Learning ◦ Variants of neural networks ◦ MLP ◦ Conv. neural networks ◦ Recurrent neural networks ◦ Training neural networks ◦ (Stochastic) Gradient Descent • Regression vs. Classification ◦ Classification as supervised learning problem ◦ Hands-On Session • Representation Learning and Generative Models ◦ AutoEncoders ◦ Directed Generative Models ◦ Undirected Generative Models ◦ Generative Adversarial Neural Networks • Probabilistic Graphical Models ◦ Bayesian Networks ◦ Variational inference (variational autoencoder)
Literature	Lecture notes as well as current literature announced in the lecture.

Course L3008: Machine Learning Applications in Electric Power Systems	
Typ	Lecture
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Christian Becker, Dr. Davood Babazadeh
Language	EN
Cycle	SoSe
Content	This part of the course focuses on how to utilize ML methods to model and operate electric power systems. Electric power systems consist of generation units such as PV, loads or consumers and the grid that connects those actors and supports to transport energy. This part of the course helps to understand the data-driven modelling of generation units (e.g. PV & fuel cells), modelling of load behavior, and to formulate and solve a state estimation problem for distribution grids using neural networks. This part of the course includes lectures to introduce the basics that are followed by practical examples and coding.
Literature	Lecture notes as well as current literature announced in the lecture.

Course L3006: Machine Learning in Electromagnetic Compatibility (EMC) Engineering	
Typ	Lecture
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Christian Schuster, Dr. Cheng Yang
Language	EN
Cycle	SoSe
Content	Electromagnetic Compatibility (EMC) Engineering deals with design, simulation, measurement, and certification of electronic and electric components and systems in such a way that their operation is safe, reliable, and efficient in any possible application. Safety is hereby understood as safe with respect to parasitic effects of electromagnetic fields on humans as well as on the operation of other components and systems nearby. Examples for components and systems range from the wiring in aircraft and ships to high-speed interconnects in server systems and wireless interfaces for brain implants. In this part of the course we will give an introduction to the physical basics of EMC engineering and then show how methods of Machine Learning (ML) can be applied to expand today's physics-based approaches in EMC Engineering.
Literature	Lecture notes as well as current literature announced in the lecture.

Course L3007: Machine Learning in High-Frequency Technology and Radar	
Typ	Lecture
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Alexander Kölpin
Language	EN
Cycle	SoSe
Content	Modern high-frequency systems benefit massively from machine learning methods. In applications where rule-based algorithms reach their limits, these data-driven approaches enable a significant increase in resolution and accuracy. This is exemplified by current research challenges, namely for the classification of targets in autonomous driving radar systems, radar-based gesture recognition for smart home applications and device control as well as in the field of medical technology for the contactless monitoring of human vital signs.
Literature	Lecture notes as well as current literature announced in the lecture.

Course L3005: Machine Learning in Wireless Communications	
Typ	Lecture
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Dr. Maximilian Stark
Language	EN
Cycle	SoSe
Content	- Supervised Learning Application - Channel Coding - Recap channel coding and block codes - Block codes as trainable neural networks - Tanner graph with trainable weights - Hands-on session - Supervised Learning Application - Modulation Detection - Recap wireless modulation schemes - Convolutional neuronal networks for blind detection of modulation schemes - Hands-on session - Autoencoder Application - Constellation Shaping I - Recap channel capacity and constellation shaping, - Capacity achieving machine learning systems - Information theoretical explanation of the autoencoder training - Hands-on session - Autoencoder Application - Constellation Shaping II - Training without a channel model - Mutual information neural estimator - Hands-on session - Generative Adversarial Network Application - Channel Modelling - Recap realistic channels with non-linear hardware impairments - Training a digital twin of a realistic channel with insufficient training data - Hands-on session - Recurrent Neural Network Application - Channel prediction - Recap time-varying channel models - Recurrent neural networks for temporal prediction - Hands-on session
Literature	Lecture notes as well as current literature announced in the lecture.

Module M1794: Applied Cryptography				
Courses				
Title	Typ		Hrs/wk	CP
Applied Cryptography (L2954)	Integrated Lecture		3	4
Applied Cryptography (L2955)	Recitation Section (small)		1	2
Module Responsible	Prof. Sibylle Fröschle			
Admission Requirements	None			
Recommended Previous Knowledge	Basics in IT security including cryptography, basics in theoretical computer science including computability and logic			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The students know and can explain - the core cryptographic primitives and constructions in the rigorous treatment of modern cryptography - cryptographic protocols for authentication and key establishment in the modern rigorous treatment - advanced key management techniques using trust anchors such as hardware security modules - the impact of quantum computers on cryptography and the advance towards post-quantum cryptography			
<i>Skills</i>	The students are able to - understand subtle attacks on inadequately applied cryptography and how to avoid them - follow security standards to develop secure cryptographic applications - systematically analyze and solve complex problems in the area of cryptographic protocols and key management - recognize novel challenges and limitations, e.g. posed by novel types of attacks			
Personal Competence				
<i>Social Competence</i>	The students are able to - expertly discuss when cryptographic solutions are necessary and what it takes to deploy them with experts and non-experts - champion the approach of modern cryptography with its focus on precise definitions and assumptions as essential for security			
<i>Autonomy</i>	The students are able to - follow up and critically assess current developments in cryptography and cryptographic applications including the challenge of post-quantum cryptography - master a new topic within the area by self-study and self-initiated interaction with experts and peers			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Exercises	Die Übungsaufgaben finden semesterbegleitend statt
Examination	Written exam			
Examination duration and scale	120 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory			

Course L2954: Applied Cryptography	
Typ	Integrated Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Sibylle Fröschle
Language	EN
Cycle	SoSe
Content	Cryptography plays a key role in securing the digital world we live in today. This course provides you with a deep understanding of symmetric and public-key cryptography based on the modern approach to cryptography. The modern approach requires that cryptographic constructions are provably secure relative to precise definitions and plausible assumptions. The assumptions typically stem from computability theory and algorithmic number theory. Moreover, you will get to know how cryptography can be applied to solve security problems such as secure communication over an open channel such as the Internet, and we will cover advanced aspects of practical deployment such as key management anchored in hardware security modules. We will see how the theme of precise definition and provable security carries forth in these areas, and are a must for today's security applications. Finally, we discuss the impact of the advent of quantum computers on cryptography and its application, and the current state of post-quantum cryptography.
Literature	Introduction to Modern Cryptography, Third Edition, Jonathan Katz and Jehuda Lindell, Chapman & Hall/CRC, 2021

Course L2955: Applied Cryptography	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Sibylle Fröschle
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Specialization Secure and Dependable IT Systems

Graduates of the Secure and Dependable IT Systems specialisation acquire extensive knowledge in software verification and IT security. They also have knowledge in communication networks and signal processing. They are able to apply methods and procedures required to work on secure and dependable IT systems, as well as critically examine new insights to further develop and incorporate in their work.

The Secure and Dependable IT Systems specialisation is recommended for students who already have a good mathematical foundation and basic knowledge in computer science and software development.

Module M0753: Software Verification				
Courses				
Title	Typ		Hrs/wk	CP
Software Verification (L0629)	Lecture		2	3
Software Verification (L0630)	Recitation Section (small)		2	3
Module Responsible	Prof. Sibylle Schupp			
Admission Requirements	None			
Recommended Previous Knowledge	- Automata theory and formal languages - Computational logic - Object-oriented programming, algorithms, and data structures - Functional programming or procedural programming - Concurrency			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	Students apply the major verification techniques in model checking and deductive verification. They explain in formal terms syntax and semantics of the underlying logics, and assess the expressivity of different logics as well as their limitations. They classify formal properties of software systems. They find flaws in formal arguments, arising from modeling artifacts or underspecification.			
<i>Skills</i>	Students formulate provable properties of a software system in a formal language. They develop logic-based models that properly abstract from the software under verification and, where necessary, adapt model or property. They construct proofs and property checks by hand or using tools for model checking or deductive verification, and reflect on the scope of the results. Presented with a verification problem in natural language, they select the appropriate verification technique and justify their choice.			
Personal Competence <i>Social Competence</i>	Students discuss relevant topics in class. They defend their solutions orally. They communicate in English.			
<i>Autonomy</i>	Using accompanying on-line material for self study, students can assess their level of knowledge continuously and adjust it appropriately. Working on exercise problems, they receive additional feedback. Within limits, they can set their own learning goals. Upon successful completion, students can identify and precisely formulate new problems in academic or applied research in the field of software verification. Within this field, they can conduct independent studies to acquire the necessary competencies and compile their findings in academic reports. They can devise plans to arrive at new solutions or assess existing ones.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	15 %	Exercises	
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory			

Course L0629: Software Verification	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	WiSe
Content	- Model checking (bounded model checking, CTL, LTL) - Real-time model checking (TCTL, timed automata) - Deductive verification (Hoare logic) - Tool support - Recent developments of verification techniques and applications
Literature	C. Baier and J-P. Katoen, Principles of Model Checking, MIT Press 2007. M. Huth and M. Bryan, Logic in Computer Science. Modelling and Reasoning about Systems, 2nd Edition, 2004. - Selected Research Papers

Course L0630: Software Verification	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0942: Software Security				
Courses				
Title	Typ		Hrs/wk	CP
Software Security (L1103)	Lecture		2	3
Software Security (L1104)	Recitation Section (small)		2	3
Module Responsible	Prof. Riccardo Scandariato			
Admission Requirements	None			
Recommended Previous Knowledge	Familiarity with C/C++, web programming			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Students can - name the main causes for security vulnerabilities in software - explain current methods for identifying and avoiding security vulnerabilities Students are capable of - performing a software vulnerability analysis - developing secure code			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>	None			
<i>Autonomy</i>	Students are capable of acquiring knowledge independently from professional publications, technical standards, and other sources, and are capable of applying newly acquired knowledge to new problems.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	5 %	Subject theoretical practical work	and Gruppenarbeit mit aktuellen Technologien aus dem Bereich Sicherheit
Examination	Written exam			
Examination duration and scale	120 minutes			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory			

Course L1103: Software Security	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	WiSe
Content	- Attacks exploiting character and integer representations - Buffer overruns - Vulnerabilities in memory managemet: use-after-free attacks - SQL injection - Cross-site scripting and cross-site request forgery - Side-channel attacks - Attacks in mobile applications - Testing for security; taint analysis - Type safe languages - Automatic vulnerability repair
Literature	M. Howard, D. LeBlanc: Writing Secure Code, 2nd edition, Microsoft Press (2002) G. Hoglund, G. McGraw: Exploiting Software: How to Break Code, Addison-Wesley (2004)

Course L1104: Software Security	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1397: Model Checking - Proof Engines and Algorithms				
Courses				
Title	Typ		Hrs/wk	CP
Model Checking - Proof Engines and Algorithms (L1979)	Lecture		2	3
Model Checking - Proof Engines and Algorithms (L1980)	Recitation Section (small)		2	3
Module Responsible	Prof. Görschwin Fey			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge about data structures and algorithms			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students know - algorithms and data structures for model checking, - basics of Boolean reasoning engines and - the impact of specification and modelling on the computational effort for model checking. <i>Skills</i> Students can - explain and implement algorithms and data structures for model checking, - decide whether a given problem can be solved using Boolean reasoning or model checking, and - implement the respective algorithms. Personal Competence <i>Social Competence</i> Students - discuss relevant topics in class and - defend their solutions orally. <i>Autonomy</i> Using accompanying material students independently learn in-depth relations between concepts explained in the lecture and additional solution strategies.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Subject theoretical practical work	andDie Aufgabe wird im Rahmen von Volresung und Prüfung definiert. Die Lösung der Aufgabe ist Zulassungsvoraussetzung für die Prüfung.
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory			

Course L1979: Model Checking - Proof Engines and Algorithms	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Görschwin Fey
Language	EN
Cycle	SoSe
Content	Correctness is a major concern in embedded systems. Model checking can fully automatically proof formal properties about digital hardware or software. Such properties are given in temporal logic, e.g., to prove "No two orthogonal traffic lights will ever be green." And how do the underlying reasoning algorithms work so effectively in practice despite a computational complexity of NP hardness and beyond? But what are the limitations of model checking? How are the models generated from a given design? The lecture will answer these questions. Open source tools will be used to gather a practical experience. Among other topics, the lecture will consider the following topics: • Modelling digital Hardware, Software, and Cyber Physical Systems • Data structures, decision procedures and proof engines ◦ Binary Decision Diagrams ◦ And-Inverter-Graphs ◦ Boolean Satisfiability ◦ Satisfiability Modulo Theories • Specification Languages ◦ CTL ◦ LTL ◦ System Verilog Assertions • Algorithms for ◦ Reachability Analysis ◦ Symbolic CTL Checking ◦ Bounded LTL-Model Checking ◦ Optimizations, e.g., induction, abstraction • Quality assurance
Literature	Edmund M. Clarke, Jr., Orna Grumberg, and Doron A. Peled. 1999. <i>Model Checking</i>. MIT Press, Cambridge, MA, USA. A. Biere, A. Biere, M. Heule, H. van Maaren, and T. Walsh. 2009. <i>Handbook of Satisfiability: Volume 185 Frontiers in Artificial Intelligence and Applications</i>. IOS Press, Amsterdam, The Netherlands, The Netherlands. Selected research papers

Course L1980: Model Checking - Proof Engines and Algorithms	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Görschwin Fey
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1773: Cybersecurity Data Science				
Courses				
Title		Typ	Hrs/wk	CP
Cybersecurity Data Science (L2914)		Lecture	2	3
Cybersecurity Data Science (L2915)		Project-/problem-based Learning	2	3
Module Responsible	Prof. Riccardo Scandariato			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge of probabilities and statistics. Familiarity with object oriented programming.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Students can: • Apply data science methods to the resolution of complex cybersecurity problems. • Use of data science methods to quantify risks and optimize cybersecurity operations. • Identify strengths and limitations of state-of-the-art methods. • Select the performance indicators of data-oriented cybersecurity solutions. • Understand cybersecurity threats in data science methods. Implement and evaluate data-driven models for the identification, treatment, and mitigation of cybersecurity risks			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>	None Students can apply the knowledge acquired throughout the course to the resolution of industrial case studies. Students should also be capable to acquire new knowledge independently from academic publications, techical standards, and white papers.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	5 %	Subject practical work	theoretical andGruppenarbeit mit aktuellen Technologien aus dem Bereich Sicherheit
Examination	Written exam			
Examination duration and scale	120 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory			

Course L2914: Cybersecurity Data Science	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	SoSe
Content	Theoretical Foundations: • Introduction to data science • Supervised and unsupervised learning • Data science methods (e.g., clustering, decision trees, artificial neural networks) • Performance indicators • Other data science methods for cybersecurity (e.g., experiments, surveys) Cybersecutrity Applications: • Spam/Phishing detection • Intrusion detection • Vulnerability/malware prediction • Adversarial machine learning
Literature	Sarker, I.H., Kayes, A.S.M., Badsha, S., Alqahtani, H., Watters, P. and Ng, A., 2020. Cybersecurity data science: an overview from machine learning perspective. Journal of Big data, 7(1) Torres, J.M., Comesaña, C.I. and Garcia-Nieto, P.J., 2019. Machine learning techniques applied to cybersecurity. International Journal of Machine Learning and Cybernetics, 10(10) Arp, D., Quiring, E., Pendlebury, F., Warnecke, A., Pierazzi, F., Wressnegger, C., Cavallaro, L. and Rieck, K., Dos and Don'ts of Machine Learning in Computer Security.

Course L2915: Cybersecurity Data Science	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1400: Design of Dependable Systems				
Courses				
Title	Typ		Hrs/wk	CP
Designing Dependable Systems (L2000)	Lecture		2	3
Designing Dependable Systems (L2001)	Recitation Section (small)		2	3
Module Responsible	Prof. Görschwin Fey			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge about data structures and algorithms			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	In the following "dependable" summarizes the concepts Reliability, Availability, Maintainability, Safety and Security. Knowledge about approaches for designing dependable systems, e.g., • Structural solutions like modular redundancy • Algorithmic solutions like handling byzantine faults or checkpointing Knowledge about methods for the analysis of dependable systems <i>Skills</i> Ability to implement dependable systems using the above approaches. Ability to analyze the dependability of systems using the above methods for analysis. Personal Competence <i>Social Competence</i> Students • discuss relevant topics in class and • present their solutions orally. <i>Autonomy</i> Using accompanying material students independently learn in-depth relations between concepts explained in the lecture and additional solution strategies.			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Subject theoretical and practical work	Die Lösung einer Aufgabe ist Zulassungsvoraussetzung für die Prüfung. Die Aufgabe wird in Vorlesung und Übung definiert.
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Embedded Systems: Elective Compulsory			

Course L2000: Designing Dependable Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Görschwin Fey
Language	EN
Cycle	SoSe
Content	Description The term dependability comprises various aspects of a system. These are typically: • Reliability • Availability • Maintainability • Safety • Security This makes dependability a core aspect that has to be considered early in system design, no matter whether software, embedded systems or full scale cyber-physical systems are considered. Contents The module introduces the basic concepts for the design and the analysis of dependable systems. Design examples for getting practical hands-on-experience in dependable design techniques. The module focuses towards embedded systems. The following topics are covered: • Modelling • Fault Tolerance • Design Concepts • Analysis Techniques
Literature	The lecture is based on a selection of research papers that is regularly updated. The specific literature references are provided with the lecture slides.

Course L2001: Designing Dependable Systems	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Görschwin Fey
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1685: Selected Aspects in Computer Science			
Courses			
Title	Typ	Hrs/wk	CP
Selected Aspects in Computer Science (L2672)	Lecture	3	4
Selected Aspects in Computer Science (L2673)	Recitation Section (small)	1	2
Module Responsible	Prof. Görschwin Fey		
Admission Requirements	None		
Recommended Previous Knowledge			
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i> <i>Skills</i> Personal Competence <i>Social Competence</i> <i>Autonomy</i>			
Workload in Hours	Depends on choice of courses		
Credit points	6		
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory		

Course L2672: Selected Aspects in Computer Science	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Examination Form	Klausur
Examination duration and scale	90 min
Lecturer	Dozenten des SD E
Language	DE/EN
Cycle	WiSe/SoSe
Content	
Literature	

Course L2673: Selected Aspects in Computer Science	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Examination Form	Fachtheoretisch-fachpraktische Arbeit
Examination duration and scale	x
Lecturer	Dozenten des SD E
Language	DE/EN
Cycle	WiSe/SoSe
Content	See interlocking course
Literature	See interlocking course

Module M2116: Advanced Seminar EIM II				
Courses				
Title		Typ	Hrs/wk	CP
Advanced Seminar EIM II (L3386)		Seminar	2	3
Module Responsible	Dozenten des SD E			
Admission Requirements	None			
Recommended Previous Knowledge	Basic knowledge in the area of the seminar within the fields of computer science, communication technology, electrical engineering or mathematics at the Master's level.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Knowledge The students are able to - explicate a specific topic from the field of the seminar, - describe complex issues, - present different views and evaluate in a critical way. Skills The students are able to - familiarize in a specific topic in limited time, - realize a literature survey on the specific topic and cite in a correct way, - elaborate a presentation and give a lecture to a selected audience, - sum up the presentation in 10-15 lines, - answer questions in the final discussion.			
Personal Competence				
Social Competence				
Autonomy				
Workload in Hours	Depends on choice of courses			
Credit points	3			
Assignment for the Following Curricula	Computer Science: Specialisation IV. Subject Specific Focus: Elective Compulsory Computer Science in Engineering: Specialisation IV. Subject Specific Focus: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory			

Course L3386: Advanced Seminar EIM II	
Typ	Seminar
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Fachtheoretisch-fachpraktische Arbeit
Examination duration and scale	x
Lecturer	Dozenten des SD E
Language	EN
Cycle	WiSe/SoSe
Content	- Seminar presentations by enrolled students about selected topics of computer science, communication technology, electrical engineering or mathematics - Active participation in discussions
Literature	Wird vom Veranstalter bekanntgegeben.

Module M2115: Advanced Seminar EIM I			
Courses			
Title	Typ	Hrs/wk	CP
Advanced Seminar EIM I (L3385)	Seminar	2	3
Module Responsible	Dozenten des SD E		
Admission Requirements	None		
Recommended Previous Knowledge	Basic knowledge in the area of the seminar within the fields of computer science, communication technology, electrical engineering or mathematics at the Master's level.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> The students are able to • explicate a specific topic from the field of the seminar, • describe complex issues, • present different views and evaluate in a critical way. <i>Skills</i> The students are able to • familiarize in a specific topic in limited time, • realize a literature survey on the specific topic and cite in a correct way, • elaborate a presentation and give a lecture to a selected audience, • sum up the presentation in 10-15 lines, • answer questions in the final discussion. Personal Competence <i>Social Competence</i> The students are able to • elaborate and introduce a topic for a certain audience, • discuss the topic, content and structure of the presentation with the instructor, • discuss certain aspects with the audience, and • as the lecturer listen and respond to questions from the audience. <i>Autonomy</i> The students are able to • define the task in question in an autonomous way, • develop the necessary knowledge, • use appropriate work equipment, and • guided by an instructor critically check the working status.		
Workload in Hours	Depends on choice of courses		
Credit points	3		
Assignment for the Following Curricula	Computer Science: Specialisation IV. Subject Specific Focus: Elective Compulsory Data Science: Core Qualification: Compulsory Electrical Engineering and Information Technology: Specialisation Microwave Engineering, Optics, and Electromagnetic Compatibility: Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Compulsory Electrical Engineering and Information Technology: Specialisation Nanoelectronics and Microsystems Technology: Compulsory Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Compulsory Computer Science in Engineering: Specialisation IV. Subject Specific Focus: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory		

Course L3385: Advanced Seminar EIM I	
Typ	Seminar
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Fachtheoretisch-fachpraktische Arbeit
Examination duration and scale	x
Lecturer	Dozenten des SD E
Language	EN
Cycle	WiSe/SoSe
Content	- Seminar presentations by enrolled students about selected topics of computer science, communication technology, electrical engineering or mathematics - Active participation in discussions
Literature	Wird vom Veranstalter bekanntgegeben.

Focus Networks

Module M0836: Communication Networks

Courses

Title	Typ	Hrs/wk	CP
Selected Topics of Communication Networks (L0899)	Project-/problem-based Learning	2	2
Communication Networks (L0897)	Lecture	2	2
Communication Networks Exercise (L0898)	Project-/problem-based Learning	1	2

Module Responsible	Dr. Koojana Kuladinithi
Admission Requirements	None
Recommended Previous Knowledge	- Fundamental stochastics - Basic understanding of computer networks and/or communication technologies is beneficial
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence <i>Knowledge</i> <i>Skills</i>	Students are able to describe the principles and structures of communication networks in detail. They can explain the formal description methods of communication networks and their protocols. They are able to explain how current and complex communication networks work and describe the current research in these examples. Students are able to evaluate the performance of communication networks using the learned methods. They are able to work out problems themselves and apply the learned methods. They can apply what they have learned autonomously on further and new communication networks.
Personal Competence <i>Social Competence</i> <i>Autonomy</i>	Students are able to define tasks themselves in small teams and solve these problems together using the learned methods. They can present the obtained results. They are able to discuss and critically analyse the solutions. Students are able to obtain the necessary expert knowledge for understanding the functionality and performance capabilities of new communication networks independently.
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70
Credit points	6
Course achievement	None
Examination	Presentation
Examination duration and scale	1.5 hours colloquium with three students, therefore about 30 min per student. Topics of the colloquium are the posters from the previous poster session and the topics of the module.
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory

Course L0899: Selected Topics of Communication Networks

Typ	Project-/problem-based Learning
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Koojana Kuladinithi
Language	EN
Cycle	WiSe
Content	Example networks selected by the students will be researched on in a PBL course by the students in groups and will be presented in a poster session at the end of the term.
Literature	see lecture

Course L0897: Communication Networks	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Koojana Kuladinithi
Language	EN
Cycle	WiSe
Content	consisting of 3 modules. - Lecture on the analysis of communication networks - Introduction to Communication Networks - Evolution of PSTN and Blocking Probability - TCP/IP Based Networks and Mobile Cellular Networks - Point-to-Point Communication - Internet - Part 1: Network Layer & Part 2: Transport Layer - Local Area Networks - Ethernet - Wireless Local Area Networks - PBL Exercises of communication networks - Performance Metrics - Trial & Error vs. Engineering Approach - Reliable Medium Access Control - Mini poster: Fast and Reliable End-to-End Data Transmission - Selected topics in communication networks This is organised as a Poster presentation of the PBL5 done for the examination. The students are asked to work on a selected topic in communication networks. This topic is selected based on the current related topic.
Literature	The students should have the basic knowledge of communication networks. Lecture 1 explains some terminologies in brief. The following book references are recommended. - Keith W. Ross James F. Kurose. Computer Networking - A Top-Down Approach. 7th ed. Pearson, 2017. A. S. Tanenbaum. Computer Networks. 5th ed. Prentice Hall, 2011. - Schiller: Mobile Communications, 8th edition. - Internet, Wikipedia

Course L0898: Communication Networks Exercise	
Typ	Project-/problem-based Learning
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Dr. Koojana Kuladinithi
Language	EN
Cycle	WiSe
Content	Part of the content of the lecture Communication Networks are reflected in computing tasks in groups, others are motivated and addressed in the form of a PBL exercise.
Literature	announced during lecture

Module M0676: Digital Communications				
Courses				
Title	Typ		Hrs/wk	CP
Digital Communications (L0444)	Lecture		2	3
Digital Communications (L0445)	Recitation Section (large)		2	2
Laboratory Digital Communications (L0646)	Practical Course		1	1
Module Responsible	Prof. Gerhard Bauch			
Admission Requirements	None			
Recommended Previous Knowledge	- Mathematics 1-3 - Signals and Systems - Fundamentals of Communications and Random Processes			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students are able to understand, compare and design modern digital information transmission schemes. They are familiar with the properties of linear and non-linear digital modulation methods. They can describe distortions caused by transmission channels and design and evaluate detectors including channel estimation and equalization. They know the principles of single carrier transmission and multi-carrier transmission as well as the fundamentals of basic multiple access schemes. The students are familiar with the contents of lecture and tutorials. They can explain and apply them to new problems. <i>Skills</i> The students are able to design and analyse a digital information transmission scheme including multiple access. They are able to choose a digital modulation scheme taking into account transmission rate, required bandwidth, error probability, and further signal properties. They can design an appropriate detector including channel estimation and equalization taking into account performance and complexity properties of suboptimum solutions. They are able to set parameters of a single carrier or multi carrier transmission scheme and trade the properties of both approaches against each other. Personal Competence <i>Social Competence</i> The students can jointly solve specific problems. <i>Autonomy</i> The students are able to acquire relevant information from appropriate literature sources. They can control their level of knowledge during the lecture period by solving tutorial problems, software tools, clicker system.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	Yes	None	Written elaboration	
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Data Science: Specialisation II. Computer Science: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Electrical Engineering and Information Technology: Core Qualification: Compulsory Electrical Engineering: Core Qualification: Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory International Management and Engineering: Specialisation II. Electrical Engineering: Elective Compulsory Microelectronics and Microsystems: Core Qualification: Elective Compulsory			

Course L0444: Digital Communications	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	WiSe
Content	- Repetition: Baseband Transmission - Pulse shaping: Non-return to zero (NRZ) rectangular pulses, raised-cosine pulses, square-root raised-cosine pulses - Power spectral density (psd) of baseband signals - Intersymbol interference (ISI) - First and second Nyquist criterion - AWGN channel - Matched filter - Matched-filter receiver and correlation receiver - Noise whitening matched filter - Discrete-time AWGN channel model - Representation of bandpass signals and systems in the equivalent baseband - Quadrature amplitude modulation (QAM) - Equivalent baseband signal and system - Analytical signal

- Equivalent baseband random process, equivalent baseband white Gaussian noise process
- Equivalent baseband AWGN channel
- Equivalent baseband channel model with frequency-offset and phase noise
- Equivalent baseband Rayleigh fading and Rice fading channel models
- Equivalent baseband frequency-selective channel model
- Discrete memoryless channels (DMC)
- Bandpass transmission via carrier modulation
 - Amplitude modulation, frequency modulation, phase modulation
 - Linear digital modulation methods
 - On-off keying, M-ary amplitude shift keying (M-ASK), M-ary phase shift keying (M-PSK), M-ary quadrature amplitude modulation (M-QAM), offset-QPSK
 - Signal space representation of transmit signal constellations and signals
 - Energy of linear digital modulated signals, average energy per symbol
 - Power spectral density of linear digital modulated signals
 - Bandwidth efficiency
 - Correlation coefficient of elementary signals
 - Error probabilities of linear digital modulation methods
 - Error functions
 - Gray mapping and natural mapping
 - Bit error probabilities, symbol error probabilities, pairwise symbol error probabilities
 - Euclidean distance and Hamming distance
 - Exact and approximate computation of error probabilities
 - Performance comparison of modulation schemes in terms of per bit SNR vs. per symbol SNR
 - Hierarchical modulation, multilevel modulation
 - Effects of carrier phase offset and carrier frequency offset
 - Differential modulation
 - M-ary differential phase shift keying (M-PSK)
 - Coherent and non-coherent detection of DPSK
 - p/M-differential phase shift keying (p/M-DPSK)
 - Differential amplitude and phase shift keying (DAPSK)
 - Non-linear digital modulation methods
 - Frequency shift keying (FSK)
 - Modulation index
 - Minimum shift keying (MSK)
 - Offset-QPSK representation of MSK
 - MSK with differential precoding and rotation
 - Bit error probabilities of MSK
 - Gaussian minimum shift keying (GMSK)
 - Power spectral density of MSK and GMSK
 - Continuous phase modulation (CPM)
 - General description of CPM signals
 - Frequency pulses and phase pulses
 - Coherent and non-coherent detection of FSK
 - Performance comparison of linear and non-linear digital modulation methods
- Frequency-selective channels, ISI channels
 - Intersymbol interference and frequency-selectivity
 - RMS delay spread
 - Narrowband and broadband channels
 - Equivalent baseband transmission model for frequency-selective channels
 - Receive filter design
- Equalization
 - Symbol-spaced and fractionally-spaced equalizers
 - Inverse system
 - Non-recursive linear equalizers
 - Linear zero-forcing (ZF) equalizer
 - Linear minimum mean squared error (MMSE) equalizer
 - Non-linear equalization:
 - Decision feedback equalizer (DFE)
 - Tomlinson-Harashima precoding
 - Maximum a posteriori probability (MAP) and maximum likelihood equalizer, Viterbi algorithm
- Single-carrier vs. multi-carrier transmission
- Multi-carrier transmission
 - General multicarrier transmission
 - Orthogonal frequency division multiplex (OFDM)
 - OFDM implementation using the Fast Fourier Transform (FFT)
 - Cyclic guard interval
 - Power spectral density of OFDM
 - Peak-to-average power ratio (PAPR)
- Multiple access
 - Principles of time division multiple access (TDMA), frequency division multiple access (FDMA), code division multiple access (CDMA), non-orthogonal multiple access (NOMA), hybrid multiple access
- Spread spectrum communications
 - Direct sequence spread spectrum communications
 - Frequency hopping
 - Protection against eavesdropping

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	◦ Protection against narrowband jammers ◦ Short vs. long spreading codes ◦ Direct sequence spread spectrum communications in frequency-selective channels ▪ Rake receiver ◦ Code division multiple access (CDMA) ▪ Design criteria of spreading sequences, autocorrelation function and crosscorrelation function of spreading sequences ▪ Intersymbol interference (ISI) and multiple access interference (MAI) ▪ Pseudo noise (PN) sequences, maximum length sequences (m-sequences), Gold codes, Walsh-Hadamard codes, orthogonal variable spreading factor (OVSF) codes ▪ Multicode transmission ▪ CDMA in uplink and downlink of a wireless communications system ▪ Single-user detection vs. multi-user detection
Literature	K. Kammeyer: Nachrichtenübertragung, Teubner P.A. Höher: Grundlagen der digitalen Informationsübertragung, Teubner. J.G. Proakis, M. Salehi: Digital Communications. McGraw-Hill. S. Haykin: Communication Systems. Wiley R.G. Gallager: Principles of Digital Communication. Cambridge A. Goldsmith: Wireless Communication. Cambridge. D. Tse, P. Viswanath: Fundamentals of Wireless Communication. Cambridge.

Course L0445: Digital Communications	
Typ	Recitation Section (large)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L0646: Laboratory Digital Communications	
Typ	Practical Course
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	WiSe
Content	- DSL transmission - Random processes - Digital data transmission
Literature	K. Kammeyer: Nachrichtenübertragung, Teubner P.A. Höher: Grundlagen der digitalen Informationsübertragung, Teubner. J.G. Proakis, M. Salehi: Digital Communications. McGraw-Hill. S. Haykin: Communication Systems. Wiley R.G. Gallager: Principles of Digital Communication. Cambridge A. Goldsmith: Wireless Communication. Cambridge. D. Tse, P. Viswanath: Fundamentals of Wireless Communication. Cambridge.

Module M0837: Simulation of Communication Networks								
Courses								
Title		Typ	Hrs/wk	CP				
Simulation of Communication Networks (L0887)		Project-/problem-based Learning	3	4				
Simulation of Communication Networks (L3476)		Lecture	2	2				
Module Responsible	Dr. Koojana Kuladinithi							
Admission Requirements	None							
Recommended Previous Knowledge	• Knowledge of computer and communication networks • Basic programming skills							
Educational Objectives	After taking part successfully, students have reached the following learning results							
Professional Competence	Knowledge Students are able to explain the necessary stochastics, the discrete event simulation technology and modelling of networks for performance evaluation. Skills Students are able to apply the method of simulation for performance evaluation to different, also not practiced, problems of communication networks. The students can analyse the obtained results and explain the effects observed in the network. They are able to question their own results. Personal Competence Social Competence Students are able to acquire expert knowledge in groups, present the results, and discuss solution approaches and results. They are able to work out solutions for new problems in small teams. Autonomy Students are able to transfer independently and in discussion with others the acquired method and expert knowledge to new problems. They can identify missing knowledge and acquire this knowledge independently.							
Workload in Hours					Independent Study Time 110, Study Time in Lecture 70			
Credit points					6			
Course achievement					None			
Examination					Oral exam			
Examination duration and scale	30 min							
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory							

Course L0887: Simulation of Communication Networks	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Dr. Koojana Kuladinithi
Language	EN
Cycle	SoSe
Content	In the course necessary basic stochastics and the discrete event simulation are introduced. Also simulation models for communication networks, for example, traffic models, mobility models and radio channel models are presented in the lecture. Students work with a simulation tool, where they can directly try out the acquired skills, algorithms and models. At the end of the course increasingly complex networks and protocols are considered and their performance is determined by simulation.
Literature	Skript des Instituts für Kommunikationsnetze Further literature is announced at the beginning of the lecture.

Course L3476: Simulation of Communication Networks	
Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Dr. Koojana Kuladinithi
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1774: Advanced Internet Computing			
Courses			
Title	Typ	Hrs/wk	CP
Advanced Internet Computing (L2916)	Lecture	2	3
Advanced Internet Computing (L2917)	Project-/problem-based Learning	2	3
Module Responsible	Prof. Stefan Schulte		
Admission Requirements	None		
Recommended Previous Knowledge	Good programming skills are necessary. Previous knowledge in the field of distributed systems is helpful.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	After successful completion of the course, students are able to: - Describe basic concepts of cloud computing and blockchain technologies - Discuss and assess critical aspects of cloud computing and blockchain technologies - Select and apply cloud and blockchain technologies for particular application areas - Design and develop practical solutions based on cloud and blockchain software The students acquire the ability to model Internet-based distributed systems and to work with these systems. This comprises especially the ability to select and utilize fitting technologies for different application areas. Furthermore, students are able to critically assess the chosen technologies. Students can work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve the problem. Students are able to independently investigate a complex problem and assess which competencies are required to solve it.		
<i>Knowledge</i>			
<i>Skills</i>			
Personal Competence			
<i>Social Competence</i>			
<i>Autonomy</i>			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56		
Credit points	6		
Course achievement	None		
Examination	Subject theoretical and practical work		
Examination duration and scale	Group project incl. presentation (50 %), written exam (60 min, 50 %)		
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory		

Course L2916: Advanced Internet Computing	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Stefan Schulte
Language	EN
Cycle	SoSe
Content	This lecture discusses modern Internet-based distributed systems in two blocks: (i) cloud computing and (ii) blockchain technologies. The following topics will be covered in the single lectures: - Cloud Service and Deployment Models - Virtualization - Further Aspects of Cloud Computing - Edge and Fog Computing - Blockchain technologies - Consensus
Literature	Lecture notes as well as current literature announced in the lecture.

Course L2917: Advanced Internet Computing	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Stefan Schulte
Language	EN
Cycle	SoSe
Content	This project-/problem-oriented part of the module augments the theoretical content of the lecture by a concrete technical problem, which needs to be solved by the students in group work during the semester. Possible topics are (blockchain-based) sensor data integration and blockchain-based voting.
Literature	Lecture notes as well as current literature announced in the lecture.

Module M1794: Applied Cryptography				
Courses				
Title	Typ		Hrs/wk	CP
Applied Cryptography (L2954)	Integrated Lecture		3	4
Applied Cryptography (L2955)	Recitation Section (small)		1	2
Module Responsible	Prof. Sibylle Fröschle			
Admission Requirements	None			
Recommended Previous Knowledge	Basics in IT security including cryptography, basics in theoretical computer science including computability and logic			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence				
<i>Knowledge</i>	The students know and can explain - the core cryptographic primitives and constructions in the rigorous treatment of modern cryptography - cryptographic protocols for authentication and key establishment in the modern rigorous treatment - advanced key management techniques using trust anchors such as hardware security modules - the impact of quantum computers on cryptography and the advance towards post-quantum cryptography			
<i>Skills</i>	The students are able to - understand subtle attacks on inadequately applied cryptography and how to avoid them - follow security standards to develop secure cryptographic applications - systematically analyze and solve complex problems in the area of cryptographic protocols and key management - recognize novel challenges and limitations, e.g. posed by novel types of attacks			
Personal Competence				
<i>Social Competence</i>	The students are able to - expertly discuss when cryptographic solutions are necessary and what it takes to deploy them with experts and non-experts - champion the approach of modern cryptography with its focus on precise definitions and assumptions as essential for security			
<i>Autonomy</i>	The students are able to - follow up and critically assess current developments in cryptography and cryptographic applications including the challenge of post-quantum cryptography - master a new topic within the area by self-study and self-initiated interaction with experts and peers			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Exercises	Die Übungsaufgaben finden semesterbegleitend statt
Examination	Written exam			
Examination duration and scale	120 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory			

Course L2954: Applied Cryptography	
Typ	Integrated Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Sibylle Fröschle
Language	EN
Cycle	SoSe
Content	Cryptography plays a key role in securing the digital world we live in today. This course provides you with a deep understanding of symmetric and public-key cryptography based on the modern approach to cryptography. The modern approach requires that cryptographic constructions are provably secure relative to precise definitions and plausible assumptions. The assumptions typically stem from computability theory and algorithmic number theory. Moreover, you will get to know how cryptography can be applied to solve security problems such as secure communication over an open channel such as the Internet, and we will cover advanced aspects of practical deployment such as key management anchored in hardware security modules. We will see how the theme of precise definition and provable security carries forth in these areas, and are a must for today's security applications. Finally, we discuss the impact of the advent of quantum computers on cryptography and its application, and the current state of post-quantum cryptography.
Literature	Introduction to Modern Cryptography, Third Edition, Jonathan Katz and Jehuda Lindell, Chapman & Hall/CRC, 2021

Course L2955: Applied Cryptography	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Sibylle Fröschle
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1700: Satellite Communications and Navigation			
Courses			
Title	Typ	Hrs/wk	CP
Radio-Based Positioning and Navigation (L2711)	Lecture	2	3
Satellite Communications (L2710)	Lecture	3	3
Module Responsible	Prof. Gerhard Bauch		
Admission Requirements	None		
Recommended Previous Knowledge	The module is designed for a diverse audience, i.e. students with different background. Basic knowledge of communications engineering and signal processing are of advantage but not required. The course intends to provide the chapters on communications techniques such that on the one hand students with a communications engineering background learn additional concepts and examples (e.g. modulation and coding schemes or signal processing concepts) which have not or in a different way been treated in our other bachelor and master courses. On the other hand, students with other background shall be able to grasp the ideas but may not be able to understand in the same depth. The individual background of the students will be taken into consideration in the oral exam.		
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence	<i>Knowledge</i> The students are able to understand, compare and analyse digital satellite communications system as well as navigation techniques. They are familiar with principal ideas of the respective communications, signal processing and positioning methods. They can describe distortions and resulting limitations caused by transmission channels and hardware components. They can describe how fundamental communications and navigation techniques are applied in selected practical systems. The students are familiar with the contents of lecture and tutorials. They can explain and apply them to new problems. <i>Skills</i> The students are able to describe and analyse digital satellite communications systems and navigation systems. They are able to analyse transmission chains including link budget calculations. They are able to choose appropriate transmission technologies and system parameters for given scenarios. Personal Competence <i>Social Competence</i> The students can jointly solve specific problems. <i>Autonomy</i> The students are able to acquire relevant information from appropriate literature sources.		
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70		
Credit points	6		
Course achievement	None		
Examination	Oral exam		
Examination duration and scale	30 min		
Assignment for the Following Curricula	Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory		

Course L2711: Radio-Based Positioning and Navigation	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Gerhard Bauch, Dr. Rico Mendrzik
Language	EN
Cycle	SoSe
Content	- Information extraction from communication signals - Time-of-arrival principle - Ranging in additive white Gaussian noise (AWGN) channel - Correlation-based range estimation - Effect of multipath propagation on time-of-arrival principle - Zero-forcing range estimation in the presence of multipath - Optimum range estimation in the presence of multipath - Zero-forcing in presence of noise - Angle-of-arrival principle - Angle-of-arrival estimation in AWGN channel - Delay-and-sum estimator - Multiple Signal Classifier (MUSIC) - MUSIC-based angle-of-arrival estimation - Case study: Comparison of estimators in AWGN channels - Effect of multipath propagation on angle-of-arrival principle - Case study: Comparison of estimators in multipath channels

- Information fusion of extracted signals
 - Distance-based positioning
 - Principle of time-of-arrival positioning
 - Geometric interpretation
 - Positioning in the absence of noise
 - Linearization of the positioning problem
 - Positioning in the presence of noise
 - Optimality criteria
 - Least squares time-of-arrival positioning
 - Maximum likelihood time-of-arrival positioning
 - Interactive Matlab demo
 - Excursion: gradient descent solvers for nonlinear programs
 - Real-life positioning with embedded development board (Arduino)
 - Linearized least squares time-of-arrival positioning
 - Effect of clock offsets on distance-based positioning
 - Time-difference-of-arrival principle
 - Least squares time-difference-of-arrival positioning
 - Clock offset mitigation via two-way ranging
 - Performance limits of distance-based positioning
 - Fisher information and the Cramér-Rao lower bound
 - Fisher information in the AWGN case
 - Multi-variate Fisher information
 - Cramér-Rao lower bound for synchronized time-of-arrival positioning
 - Case study: Synchronized time-of-arrival positioning
 - Cramér-Rao lower bound for unsynchronized time-of-arrival positioning
 - Case study: Unsynchronized time-of-arrival positioning
 - Angle-based Positioning
 - Angle-of-arrival positioning principle
 - Geometric interpretation angle-of-arrival positioning principle
 - Noise-free angle-of-arrival positioning with known orientation
 - Effect of noise on angle-of-arrival positioning
 - Least squares angle-of-arrival positioning with known orientation
 - Linear least squares angle-of-arrival positioning
 - Effect of orientation uncertainty
 - Angle-difference-of-arrival positioning
 - Geometric interpretation angle difference of arrival positioning
 - Proof of angle-difference-of-arrival locus
 - Inscribed angle lemma
 - Case study: Angle-difference-of-arrival-positioning
 - Performance limits of angle-based positioning
 - Cramér-Rao lower bound for angle-of-arrival positioning with known orientation
 - Case study: Angle-of-arrival positioning with known orientation
- Information Filtering
 - Bayesian filtering
 - Principle of Bayesian filtering
 - General Problem Formulation
 - Solution to the linear Gaussian case
 - State transition in the linear Gaussian case
 - Proof of predicted posterior distribution of the Kalman filter
 - State update in the linear Gaussian case
 - Proof of marginal posterior distribution of the Kalman filter
 - Working with Gaussian random variables
 - Proof: Affine transformation
 - Proof: Marginalization
 - Proof: Conditioning
 - Kalman filter: Optimum Inference in the linear Gaussian case
 - Modeling of process noise
 - Modeling of measurement noise
 - Case study: Kalman filtering in the linear Gaussian case
 - Interactive Kalman filtering in Matlab
 - Dealing with nonlinearities in Bayesian filtering
 - Nonlinear Gaussian case
 - Extended Kalman filter
 - Proof of predicted posterior distribution of the extended Kalman filter
 - Proof of marginal posterior distribution of the extended Kalman filter
 - Example: Nonlinear state transition
 - Case study: Extended Kalman filtering
 - Practical considerations for filter design
- Satellite Navigation
 - Overview from positioning perspective
 - Earth-centered earth-fixed (ECEF) coordinate system
 - World geodetic system (WGS)
 - Satellite navigation systems
 - System-receiver clock offsets and pseudo-ranges

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	■ Unsynchronized time-of-arrival positioning revisited ◦ GPS legacy signals and ranging ■ Signal overview ■ Time-of-arrival principle revisited ■ Direct sequence spread spectrum principle ■ Short and long codes ■ Satellite signal generation ■ Carriers and codes ■ Correlation properties of codes ■ Code division multiple access in flat fading channels ■ Navigation message ◦ Velocity estimation ◦ Hands-on case study: Design of an extended Kalman filter for satellite navigation based on recorded data • Robust navigation ◦ Multipath-assisted positioning in millimeter wave multiple antenna systems ◦ Multi-sensor fusion
Literature	

Course L2710: Satellite Communications	
Typ	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	SoSe
Content	• Introduction to satellite communications ◦ What is a satellite ◦ Overview orbits, Van Allen Belt, components of a satellite ◦ Satellite services ◦ Frequency bands for satellite services ◦ International Telecommunications Union (ITU) ◦ Influence of atmospheric impairments ◦ Milestones in satellite communications • Components of a satellite communications system ◦ Ground segment ◦ Space segment ◦ Control segment • Communication links ◦ Uplink, downlink ◦ Forward link, reverse link ◦ Intersatellite links ◦ Multiple access ◦ Performance measures ■ Effective isotropic radiated power (EIRP), antenna gain, figure of merit, G/T, carrier to noise ratio ■ Signal to noise power ratio vs. carrier to noise ratio • Single beam and multibeam satellites ◦ Beam coverage ◦ Examples for beam coverage of LEO and GEO satellites (Iridium, Viasat) • Transparent vs. regenerative payload • Orbits ◦ Low earth orbit (LEO), medium earth orbit (MEO), geosynchronous and geostationary orbits (GEO), highly elliptical orbits (HEO) ◦ Favourable orbits: ■ HEO orbits with 63-64° inclination, Molnya and Tundra orbits ■ Circular LEO orbits ■ Circular MEO Orbits (Intermediate Circular Orbits (ICO)) ■ Equatorial orbits, geostationary orbit (GEO) ◦ Important aspects of LEO, MEO and GEO satellites • Kepler's laws of planetary motion • Gravitational force • Parameters of ellipses and elliptical orbits ◦ Major and minor half axis ◦ Foci ◦ Eccentricity ◦ Eccentric anomaly, mean anomaly, true anomaly ◦ Area ◦ Orbit period ◦ Perigee, apogee ◦ Distance of satellite from center of earth ◦ Construction of ellipses according to de La Hire ◦ Orbital plane in space, inclination, right ascension (longitude) of ascending node, Vernal equinox

- Newton's laws of motion
- Newton's universal law of gravitation
- Energy of satellites: Potential energy, kinetic energy, total energy
- Instantaneous speed of a satellite
- Kepler's equation
- Satellite visibility, elevation
- Required number of LEO, MEO or GEO satellites for continuous earth coverage
- Satellite altitude and distance from a point on earth
- Choice of orbits
 - LEO, HEO, GEO
 - Elliptical orbits with non-zero inclination, Molnya orbits, Tundra orbits
 - Geosynchronous orbits
 - Parameters of geosynchronous orbits
 - Circular geosynchronous orbits
 - Inclined geosynchronous orbits
 - Quasi-zenith satellite systems (QZSS)
 - Syb-synchronous circular equatorial orbits
 - Geostationary orbit
 - Parameters of the geostationary orbit
 - Visibility
 - Propagation delay
 - Applications and system examples
- Perturbations of orbits
 - Station keeping
 - Station keeping box
 - Estimation of orbit parameters
- Fundamentals of digital communications techniques
 - Components of a digital communications system
 - Principles of encryption
 - Scrambling
 - Scrambling vs. interleaving for randomization of data sequences
 - Interleaving: Block interleaver, convolutional interleaver, random interleaver
 - Digital modulation methods
 - Linear and non-linear digital modulation methods
 - Linear digital modulation methods
 - QAM modulator and demodulator
 - Pulse shaping, square-root raised-cosine pulses
 - Average power spectral density
 - Signal space constellation
 - Examples: M-ary phase shift keying (M-PSK), M-ary quadrature amplitude shift keying (M-QAM)
 - M-PSK in noisy channels
 - Bit error probabilities of M-PSK and M-QAM
 - M-PSK vs. M-QAM
 - M-ary amplitude and phase shift keying (M-APSK)
 - M-APSK vs. M-QAM
 - Differential phase shift keying (DPSK)

Error control coding (channel coding)

- Error detecting and forward error correcting (FEC) codes
- Principle of channel coding
- Data rate, code rate, Baud rate, spectral efficiency of modulation and coding schemes
- Bandwidth-power trade-off, bandwidth-limited vs. power-limited transmission
- Coding and modulation for transparent vs. regenerative payload
- Block codes and convolutional codes
- Concatenated codes
- Bit-interleaved coded modulation
- Convolutional codes
- Low density parity check (LDPC) codes, principle of message passing decoding, bit error rate performance
- Cyclic block codes
 - Examples for cyclic block codes
 - Single errors vs. block errors, cyclic block codes for burst errors
 - Generator matrix, generator polynomials
 - Systematic encoding and syndrome determination with shift registers
 - Cyclic redundancy check (CRC) codes
- Automatic repeat request (ARQ)
 - Principle of ARQ
 - Stop-and-wait ARQ
 - Go-back-N ARQ
 - Selective-repeat ARQ
- Transmission gains and losses

	◦ Antenna gain ■ Antenna radiation pattern ■ Maximum antenna gain, 3dB beamwidth ■ Maximum antenna gain of circular aperture ■ Maximum antenna gain of a geostationary satellite with global coverage ◦ Effective isotropic radiated power (EIRP) ◦ Power flux density ◦ Path loss ■ Free space loss, free space loss for geostationary satellites ■ Atmospheric loss ■ Received power ◦ Losses in transmit and receive equipment ■ Feeder loss ■ Depointing loss ■ Polarization mismatch loss ◦ Combined effect of losses • Noise ◦ Origins of noise ◦ White noise ◦ Noise power spectral density and noise power ◦ Additive white Gaussian noise (AWGN) channel model ◦ Antenna noise temperature ◦ Earth brightness temperature ◦ Signal to noise ratios • Atmospheric distortions ◦ Atmosphere of the earth: Troposphere, stratosphere, mesosphere, thermosphere, exosphere ◦ Attenuation and depolarization due to rain, fog, rain and ice clouds, sandstorms ◦ Scintillation ◦ Faraday effect ◦ Multipath contributions • Link budget calculations ◦ GEO clear sky uplink and downlink ◦ GEO uplink and downlink under rain conditions ◦ Transparent vs. regenerative payload • Link availability improvement through site diversity and adaptive transmission ◦ Transparent vs. regenerative payload ■ Non-linear amplifiers ■ Saleh model, Rapp model ■ Input and output back-off factor ■ Single carrier and multicarrier operation ■ Dimensioning of transmission parameters ■ Sources of noise: Thermal noise, interference, intermodulation products ■ Signal to noise ratio and bit error probability ■ Robustness against interference and non-linear channels • Satellite networks ◦ Satellite network reference architectures ◦ Network topologies ◦ Network connectivity ■ Types of network connectivity ■ On-board connectivity ■ Inter-satellite links ◦ Broadcast networks ◦ Satellite-based internet • Satellite communications systems and standards examples ◦ The role of standards in satellite communications ◦ The Digital Video Broadcast Satellite Standard: DVB-S, DVB-S2, DVB-S2X ◦ Satellites in 3GPP mobile communications networks ◦ LEO megaconstellations: SpaceX Starlink, Kuiper, OneWeb ◦ Space debris ◦ IRIS² (Infrastructure for Resilience, Interconnectivity and Security by Satellite): The new EU Secure Satellite Constellation ◦ The German Heinrich Hertz mission
Literature	

Module M0839: Traffic Engineering				
Courses				
Title		Typ	Hrs/wk	CP
Network Optimization of Communication Networks (L0902)		Project-/problem-based Learning	2	3
Queueing Theory for Communication Networks (L0900)		Lecture	1	2
Queueing Theory for Communication Networks (L0901)		Recitation Section (small)	1	1
Module Responsible	Prof. Andreas Timm-Giel			
Admission Requirements	None			
Recommended Previous Knowledge	- Fundamentals of communication or computer networks - Stochastics			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students are able to describe methods for planning, optimisation and performance evaluation of communication networks. <i>Skills</i> Students are able to solve typical planning and optimisation tasks for communication networks. Furthermore they are able to evaluate the network performance using queueing theory. Students are able to apply independently what they have learned to other and new problems. They can present their results in front of experts and discuss them.			
Personal Competence				
Social Competence				
Autonomy				
Students are able to acquire the necessary expert knowledge to understand the functionality and performance of new communication networks independently.				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Oral exam			
Examination duration and scale	30 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory			

Course L0902: Network Optimization of Communication Networks	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Andreas Timm-Giel, Konrad Fuger
Language	EN
Cycle	WiSe
Content	Selected applications of methods for planning, optimization, and performance evaluation of communication networks, which have been introduced in the traffic engineering lecture are prepared by the students and presented in a seminar.
Literature	U. Killat, Entwurf und Analyse von Kommunikationsnetzen, Vieweg + Teubner - further literature announced in the lecture

Course L0900: Queuing Theory for Communication Networks	
Typ	Lecture
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Andreas Timm-Giel, Konrad Fuger
Language	EN
Cycle	WiSe
Content	Network Planning and Optimization • Linear Programming (LP) • Network planning with LP solvers • Planning of communication networks Queueing Theory for Communication Networks • Stochastic processes • Queueing systems • Switches (circuit- and packet switching) • Network of queues
Literature	Literatur: U. Killat, Entwurf und Analyse von Kommunikationsnetzen, Springer Weitere Literatur wird in der Lehrveranstaltung bekanntgegeben / Literature: U. Killat, Entwurf und Analyse von Kommunikationsnetzen, Springer further literature announced in the lecture

Course L0901: Queuing Theory for Communication Networks	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Andreas Timm-Giel, Konrad Fuger
Language	EN
Cycle	WiSe
Content	Accompanying exercise for the traffic engineering course
Literature	Literatur: U. Killat, Entwurf und Analyse von Kommunikationsnetzen, Springer Weitere Literatur wird in der Lehrveranstaltung bekanntgegeben / Literature: U. Killat, Entwurf und Analyse von Kommunikationsnetzen, Springer further literature announced in the lecture

Focus Software and Signal Processing

Module M2001: Reinforcement Learning

Courses

Title	Typ	Hrs/wk	CP
Reinforcement Learning (L3305)	Lecture	2	3
Reinforcement Learning (L3306)	Project-/problem-based Learning	2	3

Module Responsible	Dr. Manfred Eppe
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Admission Requirements	None
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Recommended Previous Knowledge	- Lectures at TUHH: - Machine Learning I and II - Advanced Machine Learning - Programming Paradigms - Additional recommended knowledge and prerequisites: - Python programming, including libraries for neural networks, deep learning and data visualization (e.g. wandb). - Access to a decent PC or laptop with at least mediocre performance.
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Educational Objectives	After taking part successfully, students have reached the following learning results
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Professional Competence <i>Knowledge</i>	- Foundations of Reinforcement Learning, including - Markov Decision Processes - Dynamic Programming - Value- and Policy Iteration - TD-Learning and MC-Methods - Policy Gradient Methods - Q-Learning - Actor-Critic Methods - Individual advanced competences, at least one of the following: - Soft Actor-Critic - Proximal Policy Optimization - Multi-Agent RL - Goal-Conditioned RL - Intrinsic Rewards and Reward Shaping - Hierarchical RL - Model-based RL
<i>Skills</i>	- Using a research framework for scientific questions in reinforcement learning - Data visualization and empirical evaluation of research data - Systematic validation of hypotheses - Transfer of theoretical background to practical applications
Personal Competence <i>Social Competence</i>	- Working in a team - Conflict resolution in a team - Coordination of tasks in a team
<i>Autonomy</i>	- Improvement of English writing and speaking - Scientific work - Presentation skills

Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
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Credit points	6
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Course achievement	Compulsory	Bonus	Form	Description
	Yes	20 %	Presentation	Projektpräsentation

Examination	Oral exam
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Examination duration and scale	30 min
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Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory
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Course L3305: Reinforcement Learning	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Manfred Eppe, Frank Röder
Language	EN
Cycle	WiSe
Content	How can robotic and artificial agents learn to act appropriately in their environment? This course conveys foundational and advanced concepts of Reinforcement Learning to address this question. The core concepts will be taught in a lecture. In addition, there will be a strong focus on practical project work in the seminar part of this course. The practical applications will involve virtual agents in simulated 2d- or 3d environments. Prerequisites for participation are knowledge in machine learning, neural networks, stochastics, Python programming and a decent own PC or laptop with at least mediocre performance. The corresponding project/problem-based course complements the theoretical content of the lecture with a specific technical problem (mini-projects) that students solve in group work throughout the semester. The topics come from the field of robotics and are solved using 3D simulations in our institute's virtual robotics laboratory (Scilab-RL).
Literature	Reinforcement Learning: An Introduction (Sutton & Barto) Wiki and website of our virtual robotics lab (Scilab-RL): https://scilab-rl.github.io/Scilab-RL/

Course L3306: Reinforcement Learning	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Manfred Eppe, Frank Röder
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M2051: Secure Software Application Project				
Courses				
Title	Typ		Hrs/wk	CP
Secure Software Application Project (L3307)	Project-/problem-based Learning		4	6
Module Responsible	Prof. Riccardo Scandariato			
Admission Requirements	None			
Recommended Previous Knowledge	Some familiarity with web programming (e.g., Javascript, PHP). Prior attendance to the courses "Software Security" or "Secure Software Engineering" is beneficial but not mandatory.			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	Upon the completion of this modules, students will be able to:			
	- Recognize coding pitfalls that have security implications. - Select suitable methods for security assessment. - Interact and collaborate effectively in an agile team.			
<i>Skills</i>	Upon the completion of this modules, students will be able to:			
	- Conduct security evaluations of software systems. - Write secure code. - Fix vulnerable code.			
Personal Competence <i>Social Competence</i>	The project is done in groups and in close cooperation within the teammates is required. Upon the completion of this modules, students will be able to:			
	- Interact and collaborate effectively in an agile team. - Coordinate development effort. - Resolve conflicts and provide constructive feedback			
<i>Autonomy</i>	After completing the module, students will be able to independently carry out secure development software projects.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written elaboration			
Examination duration and scale	Report and presentation about the achieved results			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory			

Course L3307: Secure Software Application Project	
Typ	Project-/problem-based Learning
Hrs/wk	4
CP	6
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	WiSe/SoSe
Content	This is a hands-on project course (PBL) where students will work together in development teams (e.g. 5 students). The overall goal of the project is to: - Acquire better software development skills. - Adopt secure programming practices in a realistic context. - Incorporate security best practices in software development. A functional prototype of a realistic software application is provided (e.g., a web store written in Python and Javascript) and the teams will have to progress through some development phases, including: (i) fixing functional bugs and security vulnerabilities, (ii) extend the functionality and the security features of the system, and (iii) validate the system via state-of-the-art security assessment techniques (e.g., static security application testing, or threat analysis). Prior attendance to the courses "Software Security" or "Secure Software Engineering" is beneficial but not mandatory. Some introductory (theoretical) material will be provided in this course.
Literature	1. Greg Hoglund, Gary McGraw, Exploiting Software: How to Break Code, Addison-Wesley, 2004 2. Mark G. Graff, Kenneth R. Van Wyk, Debby Russell, Secure Coding: Principles and Practices, O'Reilly, 2003

Module M1682: Secure Software Engineering				
Courses				
Title	Typ		Hrs/wk	CP
Secure Software Engineering (L2667)	Lecture		2	3
Secure Software Engineering (L2668)	Project-/problem-based Learning		2	3
Module Responsible	Prof. Riccardo Scandariato			
Admission Requirements	None			
Recommended Previous Knowledge	Familiarity with basic software engineering concepts (e.g., requirements, design) and basic security concepts (e.g., confidentiality, integrity, availability)			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	Students can: - Elicit security requirements in a software project - Model and document security measures in a software design - Use threat and risk analysis techniques - Understand how security code reviews are performed - Understand the core definitions of concepts related to privacy - Understand privacy enhancing technologies Select appropriate security assurance techniques to be used in a security assurance program			
<i>Knowledge</i>				
<i>Skills</i>				
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>	Students can apply the knowledge acquired throughout the course to the resolution of industrial case studies. Students should also be capable to acquire new knowledge independently from academic publications, technical standards, and white papers.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	5 %	Subject theoretical and practical work	and Gruppenarbeit mit aktuellen Technologien aus dem Bereich Sicherheit
Examination	Written exam			
Examination duration and scale	120 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory			

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Course L2667: Secure Software Engineering	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	SoSe
Content	- Secure software development processes and maturity models - Techniques to define security requirements - Techniques to create, document and analyse the design of secure applications - Threat and risk analysis techniques - Security code reviews - Program repair techniques for security vulnerabilities - Privacy engineering
Literature	Charles Haley, Robin Laney, Jonathan Moffett, Bashar Nuseibeh, 2008, Security Requirements Engineering: A Framework for Representation and Analysis, Transactions on Software Engineering Sindre, G. and Opdahl, A.L., 2005. Eliciting security requirements with misuse cases. Requirements engineering, 10(1) Fontaine, P.J., Van Lamsweerde, A., Letier, E. and Darimont, R., 2001. Goal-oriented elaboration of security requirements. Joanna C. S. Santos, Katy Tarrit, Mehdi Mirakhorli, 2017, A Catalog of Security Architecture Weaknesses, IEEE International Conference on Software Architecture Workshops (ICSAW) Jürjens, J., 2002, UMLsec: Extending UML for secure systems development, International Conference on The Unified Modeling Language Lund, M.S., Solhaug, B. and Stølen, K., 2011. A guided tour of the CORAS method. In Model-Driven Risk Analysis, Springer Pratyusa Manadhata, Jeannette Wing, 2021, An Attack Surface Metric, IEEE Transactions on Software Engineering, 37(3) Howard, M.A., 2006. A process for performing security code reviews. IEEE Security & privacy, 4(4) Jaap-Henk Hoepman, 2022, Privacy Design Strategies (The Little Blue Book) Diaz, C. and Gürses, S., 2012. Understanding the landscape of privacy technologies. Proceedings of the information security summit, 12

Course L2668: Secure Software Engineering	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1301: Software Testing				
Courses				
Title	Typ		Hrs/wk	CP
Software Testing (L1791)	Lecture		2	3
Software Testing (L1792)	Project-/problem-based Learning		2	3
Module Responsible	Prof. Sibylle Schupp			
Admission Requirements	None			
Recommended Previous Knowledge	• Software Engineering • Higher Programming Languages • Object-Oriented Programming • Algorithms and Data Structures • Experience with (Small) Software Projects • Statistics			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> Students explain the different phases of testing, describe fundamental techniques of different types of testing, and paraphrase the basic principles of the corresponding test process. They give examples of software development scenarios and the corresponding test type and technique. They explain algorithms used for particular testing techniques and describe possible advantages and limitations. <i>Skills</i> Students identify the appropriate testing type and technique for a given problem. They adapt and execute respective algorithms to execute a concrete test technique properly. They interpret testing results and execute corresponding steps for proper re-test scenarios. They write and analyze test specifications. They apply bug finding techniques for non-trivial problems. Personal Competence <i>Social Competence</i> Students discuss relevant topics in class. They defend their solutions orally. They communicate in English. <i>Autonomy</i> Students can assess their level of knowledge continuously and adjust it appropriately, based on feedback and on self-guided studies. Within limits, they can : own learning goals. Upon successful completion, students can identify and precisely formulate new problems in academic or applied research in the field of : testing. Within this field, they can conduct independent studies to acquire the necessary competencies and compile their findings in academic reports. T devise plans to arrive at new solutions or assess existing ones			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Subject theoretical and practical work			
Examination duration and scale	Software			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory			

Course L1791: Software Testing	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	SoSe
Content	• Fundamentals of software testing • Model-based testing • Test automation • Criteria-based testing
Literature	• P. Ammann and J. Offutt, "Introduction to Software Testing", 2nd edition 2016. • Selected research papers

Course L1792: Software Testing	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	SoSe
Content	• Fundamentals of software testing • Model-based testing • Test automation • Criteria-based testing
Literature	• M. Pezze and M. Young, Software Testing and Analysis, John Wiley 2008. • P. Ammann and J. Offutt, "Introduction to Software Testing", 2nd edition 2015.

Module M1810: Autonomous Cyber-Physical Systems				
Courses				
Title	Typ		Hrs/wk	CP
Autonomous Cyber-Physical Systems (L3000)	Lecture		2	3
Autonomous Cyber-Physical Systems (L3001)	Project/problem-based Learning		3	3
Module Responsible	Prof. Bernd-Christian Renner			
Admission Requirements	None			
Recommended Previous Knowledge	• Very good knowledge and practical experience in programming in the C/C++ language (e.g., module: Procedural Programming for Computer Scientists) • Basic knowledge in software engineering • Basic knowledge in wired and wireless communication protocols • Principal understanding of simple electronic circuits			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence <i>Knowledge</i>	This master's course centers on the advanced Internet of Things (IoT) systems, spotlighting energy sustainability and Real-Time Operating Systems (RTOS) within cyber-physical frameworks. Key focus areas include the exploration of IoT system architecture, especially leveraging RTOS for real-time responsiveness, and energy-efficient technologies for sustainable operations. Upon completion, students will be capable of: • Outlining the structure and operational principles of IoT systems and their significance in modern cyber-physical environments. • Differentiating between RTOS and conventional operating systems in the context of IoT applications. • Evaluating energy harvesting (specifically solar technologies) and storage systems for their applicability and efficiency in IoT devices • Knowing different wireless technologies (e.g., Bluetooth LE, LoRaWAN, Zigbee) and networking protocols (TCP/IP, MQTT) for IoT connectivity and functionality, and pinpointing their dis-/advantages.			
<i>Skills</i>	Students will acquire practical skills to: • Describe, plan, develop and optimize IoT applications using RTOS, focusing on multitasking, efficient memory, and power management. • Select and deploy suitable wireless and networking technologies considering device requirements and operational contexts. • Integrate energy harvesting and storage technologies into IoT solutions, enhancing their sustainability and autonomy. • Conduct comprehensive planning and evaluation of IoT projects, incorporating scientific methodologies for real-world applications.			
Personal Competence <i>Social Competence</i>	• Students are able to work goal-oriented in small mixed groups. • They learn and broaden their teamwork abilities. • They learn to define and split tasks within the team. • After completing the module, the students are able to work on similar tasks alone or in a group and to present the results in a suitable way.			
<i>Autonomy</i>	After completing the module, the students are able to independently work on sub-areas of the subject using specialist literature, to summarize and present the knowledge they have acquired and to link it to the content of other courses.			
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Attestation	
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L3000: Autonomous Cyber-Physical Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	
Literature	

Course L3001: Autonomous Cyber-Physical Systems	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1694: Security of Cyber-Physical Systems				
Courses				
Title	Typ		Hrs/wk	CP
Security of Cyber-Physical Systems (L2691)	Integrated Lecture		3	4
Security of Cyber-Physical Systems (L2692)	Recitation Section (small)		1	2
Module Responsible	Prof. Sibylle Fröschle			
Admission Requirements	None			
Recommended Previous Knowledge	IT security, programming skills, statistics			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students know and can explain - the threats posed by cyber attacks to cyber-physical systems (CPS) - concrete attacks at a technical level, e.g. on bus systems - security solutions specific to CPS with their capabilities and limitations - examples of security architectures for CPS and the requirements they guarantee - standard security engineering processes for CPS <i>Skills</i> The students are able to - identify security threats and assess the risks for a given CPS - apply attack toolkits to analyse a networked control system, and detect attacks beyond those taught in class - identify and apply security solutions suitable to the requirements - follow security engineering processes to develop a security architecture for a given CPS - recognize challenges and limitations, e.g. posed by novel types of attack			
Personal Competence				
<i>Social Competence</i>				
<i>Autonomy</i>				
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	Compulsory	Bonus	Form	Description
	No	10 %	Exercises	Die Übungsaufgaben finden semesterbegleitend statt.
Examination	Written exam			
Examination duration and scale	120 min			
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory			

Module Manual M.Sc. "Information and Communication Systems"

Course L2691: Security of Cyber-Physical Systems	
Typ	Integrated Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Sibylle Fröschle
Language	EN
Cycle	WiSe
Content	Embedded systems in energy, production, and transportation are currently undergoing a technological transition to highly networked automated cyber-physical systems (CPS). Such systems are potentially vulnerable to cyber attacks, and these can have physical impact. In this course we investigate security threats, solutions and architectures that are specific to CPS. The topics are as follows: Fundamentals and motivating examples Networked and embedded control systems Bus system level attacks Intruder detection systems (IDS), in particular physics-based IDS System security architectures, including cryptographic solutions Adversarial machine learning attacks in the physical world Aspects of Location and Localization Wireless networks and infrastructures for critical applications Communication security architectures and remaining threats Intruder detection systems (IDS), in particular data-centric IDS Resilience against multi-instance attacks Security Engineering of CPS: Process and Norms
Literature	Recent scientific papers and reports in the public domain.

Course L2692: Security of Cyber-Physical Systems	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Sibylle Fröschle
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1598: Image Processing				
Courses				
Title	Typ		Hrs/wk	CP
Image Processing (L2443)	Lecture		2	4
Image Processing (L2444)	Recitation Section (small)		2	2
Module Responsible	Prof. Tobias Knopp			
Admission Requirements	None			
Recommended Previous Knowledge	Signal and Systems			
Educational Objectives	After taking part successfully, students have reached the following learning results			
Professional Competence	<i>Knowledge</i> The students know about • visual perception • multidimensional signal processing • sampling and sampling theorem • filtering • image enhancement • edge detection • multi-resolution procedures: Gauss and Laplace pyramid, wavelets • image compression • image segmentation • morphological image processing <i>Skills</i> The students can • analyze, process, and improve multidimensional image data • implement simple compression algorithms • design custom filters for specific applications Personal Competence <i>Social Competence</i> Students can work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve the problem. <i>Autonomy</i> Students are able to independently investigate a complex problem and assess which competencies are required to solve it.			
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56			
Credit points	6			
Course achievement	None			
Examination	Written exam			
Examination duration and scale	90 min			
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Data Science: Specialisation I. Mathematics/Computer Science: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Medical Technology: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory			

Course L2443: Image Processing	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Tobias Knopp
Language	EN
Cycle	WiSe
Content	• Visual perception • Multidimensional signal processing • Sampling and sampling theorem • Filtering • Image enhancement • Edge detection • Multi-resolution procedures: Gauss and Laplace pyramid, wavelets • Image Compression • Segmentation • Morphological image processing
Literature	Bredies/Lorenz, Mathematische Bildverarbeitung, Vieweg, 2011 Pratt, Digital Image Processing, Wiley, 2001 Bernd Jähne: Digitale Bildverarbeitung - Springer, Berlin 2005

Course L2444: Image Processing	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Tobias Knopp
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Thesis

Module M-002: Master Thesis

Courses

Title	Typ	Hrs/wk	CP
Module Responsible	Professoren der TUHH		
Admission Requirements	According to General Regulations §21 (1): At least 60 credit points have to be achieved in study programme. The examinations board decides on exceptions.		
Recommended Previous Knowledge			
Educational Objectives	After taking part successfully, students have reached the following learning results		
Professional Competence <i>Knowledge</i>	- The students can use specialized knowledge (facts, theories, and methods) of their subject competently on specialized issues. - The students can explain in depth the relevant approaches and terminologies in one or more areas of their subject, describing current developments and taking up a critical position on them. - The students can place a research task in their subject area in its context and describe and critically assess the state of research.		
Skills	The students are able: - To select, apply and, if necessary, develop further methods that are suitable for solving the specialized problem in question. - To apply knowledge they have acquired and methods they have learnt in the course of their studies to complex and/or incompletely defined problems in a solution-oriented way. - To develop new scientific findings in their subject area and subject them to a critical assessment.		
Personal Competence <i>Social Competence</i>	Students can - Both in writing and orally outline a scientific issue for an expert audience accurately, understandably and in a structured way. - Deal with issues competently in an expert discussion and answer them in a manner that is appropriate to the addressees while upholding their own assessments and viewpoints convincingly.		
Autonomy	Students are able: - To structure a project of their own in work packages and to work them off accordingly. - To work their way in depth into a largely unknown subject and to access the information required for them to do so. - To apply the techniques of scientific work comprehensively in research of their own.		
Workload in Hours	Independent Study Time 900, Study Time in Lecture 0		
Credit points	30		
Course achievement	None		
Examination	Thesis		
Examination duration and scale	According to General Regulations		
Assignment for the Following Curricula	Civil Engineering: Thesis: Compulsory Bioprocess Engineering: Thesis: Compulsory Chemical and Bioprocess Engineering: Thesis: Compulsory Chemical and Bioprocess Engineering: Thesis: Compulsory Computational Engineering: Thesis: Compulsory Computer Science: Thesis: Compulsory Data Science: Thesis: Compulsory Electrical Engineering and Information Technology: Thesis: Compulsory Electrical Engineering: Thesis: Compulsory Energy Systems: Thesis: Compulsory Environmental Engineering: Thesis: Compulsory Aircraft Systems Engineering: Thesis: Compulsory Global Innovation Management: Thesis: Compulsory Computer Science in Engineering: Thesis: Compulsory Information and Communication Systems: Thesis: Compulsory Interdisciplinary Mathematics: Thesis: Compulsory International Production Management: Thesis: Compulsory International Management and Engineering: Thesis: Compulsory Joint European Master in Environmental Studies - Cities and Sustainability: Thesis: Compulsory Logistics, Infrastructure and Mobility: Thesis: Compulsory Aeronautics: Thesis: Compulsory Mechanical Engineering - Product Development and Production: Thesis: Compulsory Materials Science and Engineering: Thesis: Compulsory		

	Materials Science: Thesis: Compulsory
	Mechanical Engineering and Management: Thesis: Compulsory
	Mechatronics: Thesis: Compulsory
	Biomedical Engineering: Thesis: Compulsory
	Microelectronics and Microsystems: Thesis: Compulsory
	Product Development, Materials and Production: Thesis: Compulsory
	Renewable Energies: Thesis: Compulsory
	Naval Architecture and Ocean Engineering: Thesis: Compulsory
	Naval Architecture and Ocean Engineering: Thesis: Compulsory
	Ship and Offshore Technology: Thesis: Compulsory
	Theoretical Mechanical Engineering: Thesis: Compulsory
	Process Engineering: Thesis: Compulsory
	Water and Environmental Engineering: Thesis: Compulsory
	Certification in Engineering & Advisory in Aviation: Thesis: Compulsory